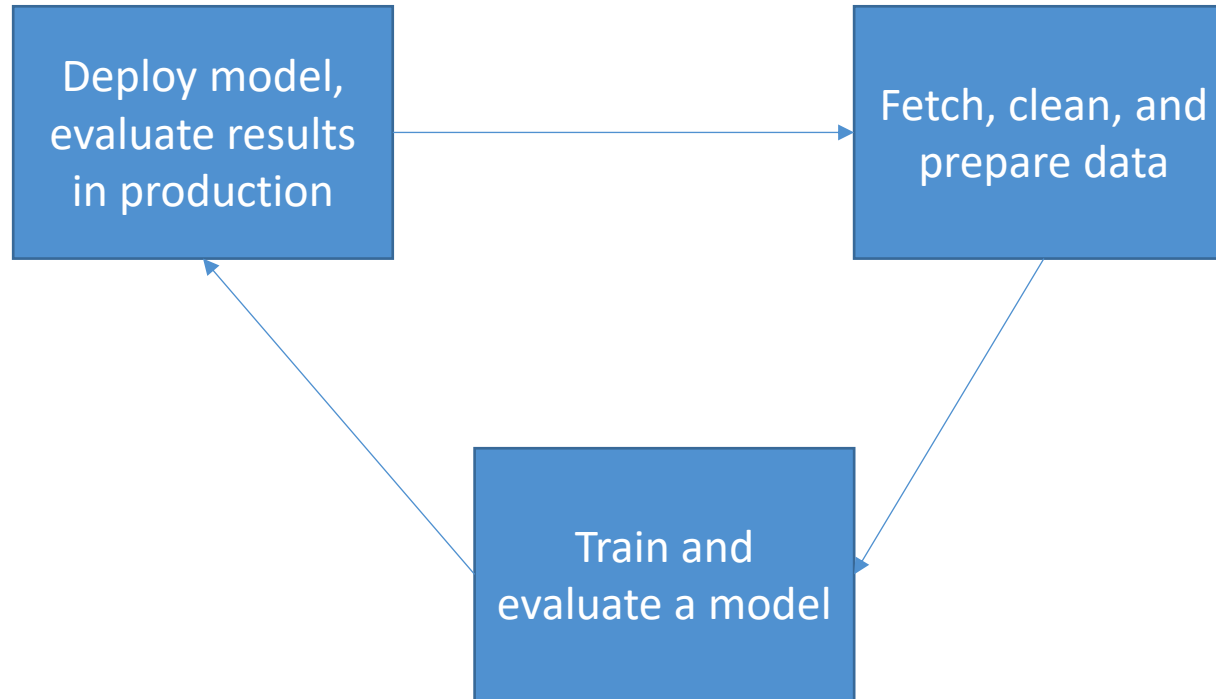
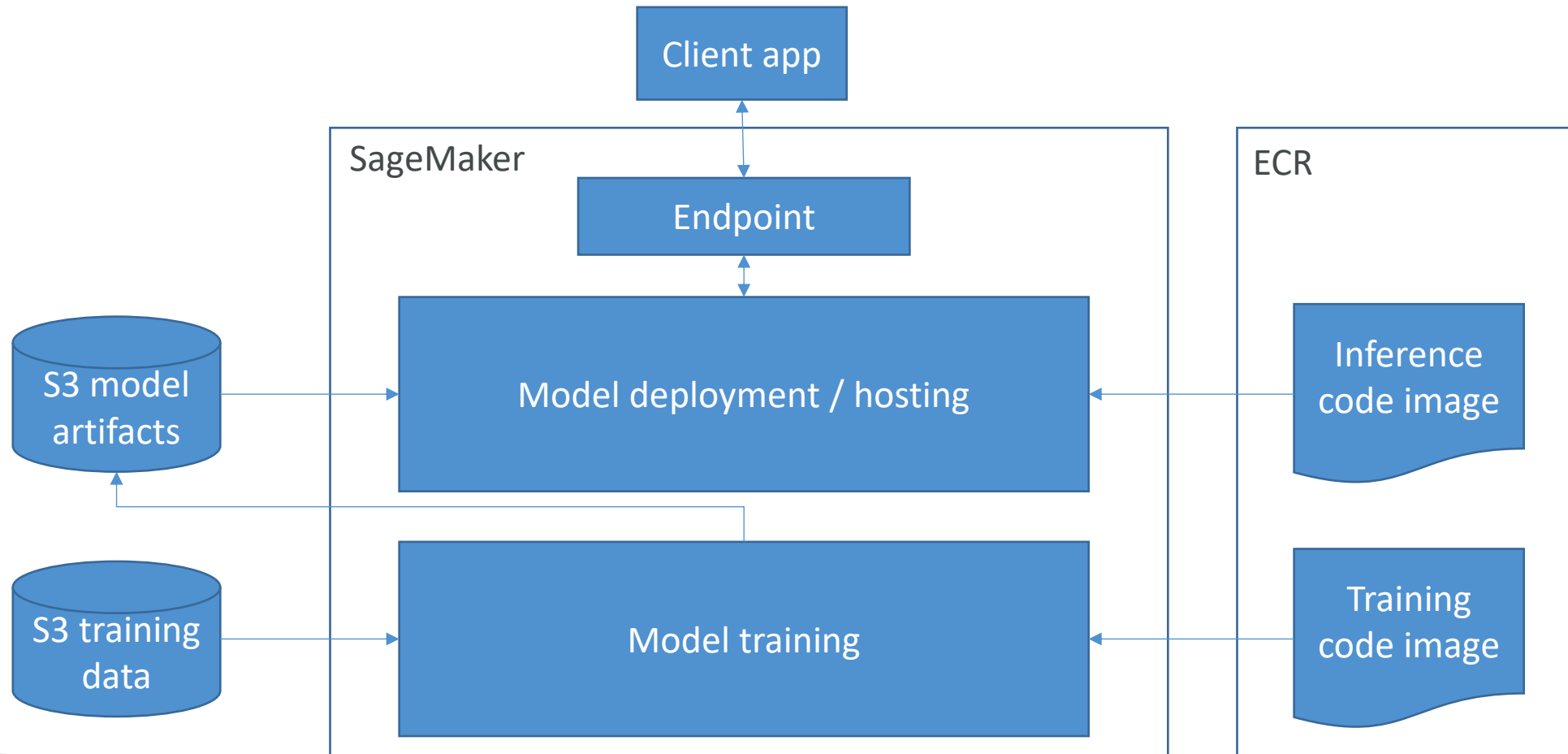


Amazon SageMaker

SageMaker is built to handle the entire machine learning workflow.

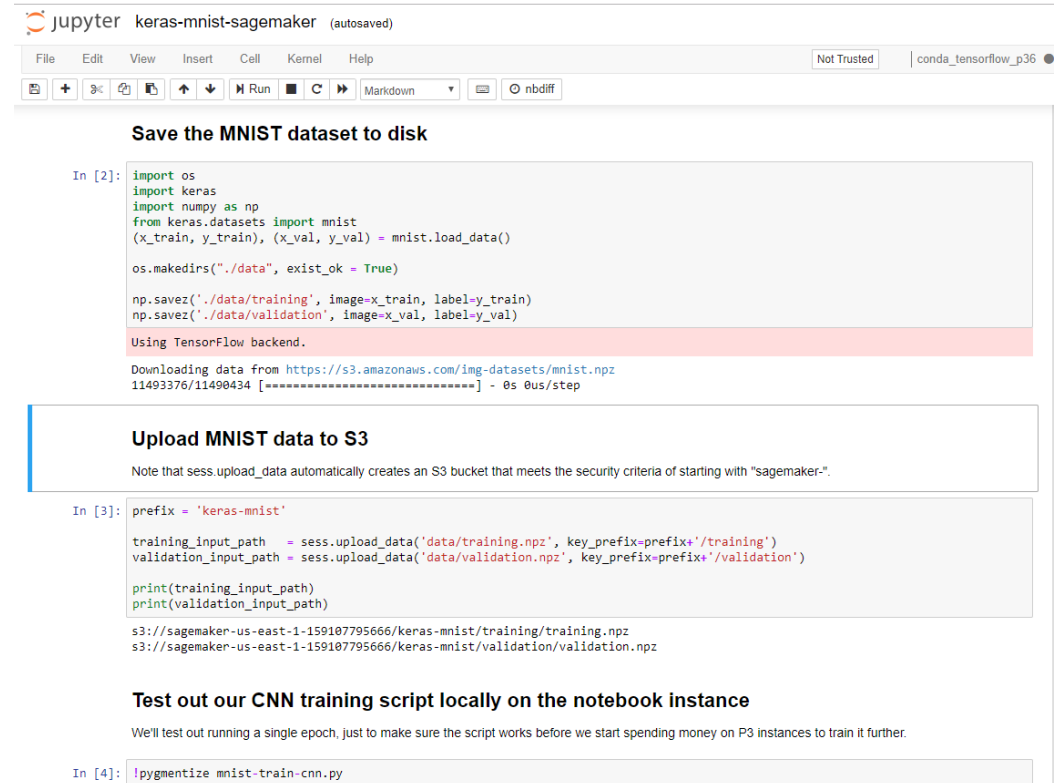


SageMaker Training & Deployment



SageMaker Notebooks can direct the process

- Notebook Instances on EC2 are spun up from the console
 - S3 data access
 - Scikit_learn, Spark, Tensorflow
 - Wide variety of built-in models
 - Ability to spin up training instances
 - Ability to deploy trained models for making predictions at scale



The screenshot shows a Jupyter Notebook interface with the title 'keras-mnist-sagemaker (autosaved)'. The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Help) and a toolbar with icons for file operations, running cells, and markdown. The notebook content is divided into sections:

- Save the MNIST dataset to disk**: This section contains a code cell (In [2]) with the following code:

```
import os
import keras
import numpy as np
from keras.datasets import mnist
(x_train, y_train), (x_val, y_val) = mnist.load_data()

os.makedirs("./data", exist_ok = True)

np.savez('./data/training', image=x_train, label=y_train)
np.savez('./data/validation', image=x_val, label=y_val)

Using TensorFlow backend.
```

Below the code, a message indicates: "Downloading data from https://s3.amazonaws.com/img-datasets/mnist.npz 11493376/11490434 [=====] - 0s 0us/step".
- Upload MNIST data to S3**: This section contains a code cell (In [3]) with the following code:

```
prefix = 'keras-mnist'

training_input_path = sess.upload_data('data/training.npz', key_prefix=prefix+'/training')
validation_input_path = sess.upload_data('data/validation.npz', key_prefix=prefix+'/validation')

print(training_input_path)
print(validation_input_path)

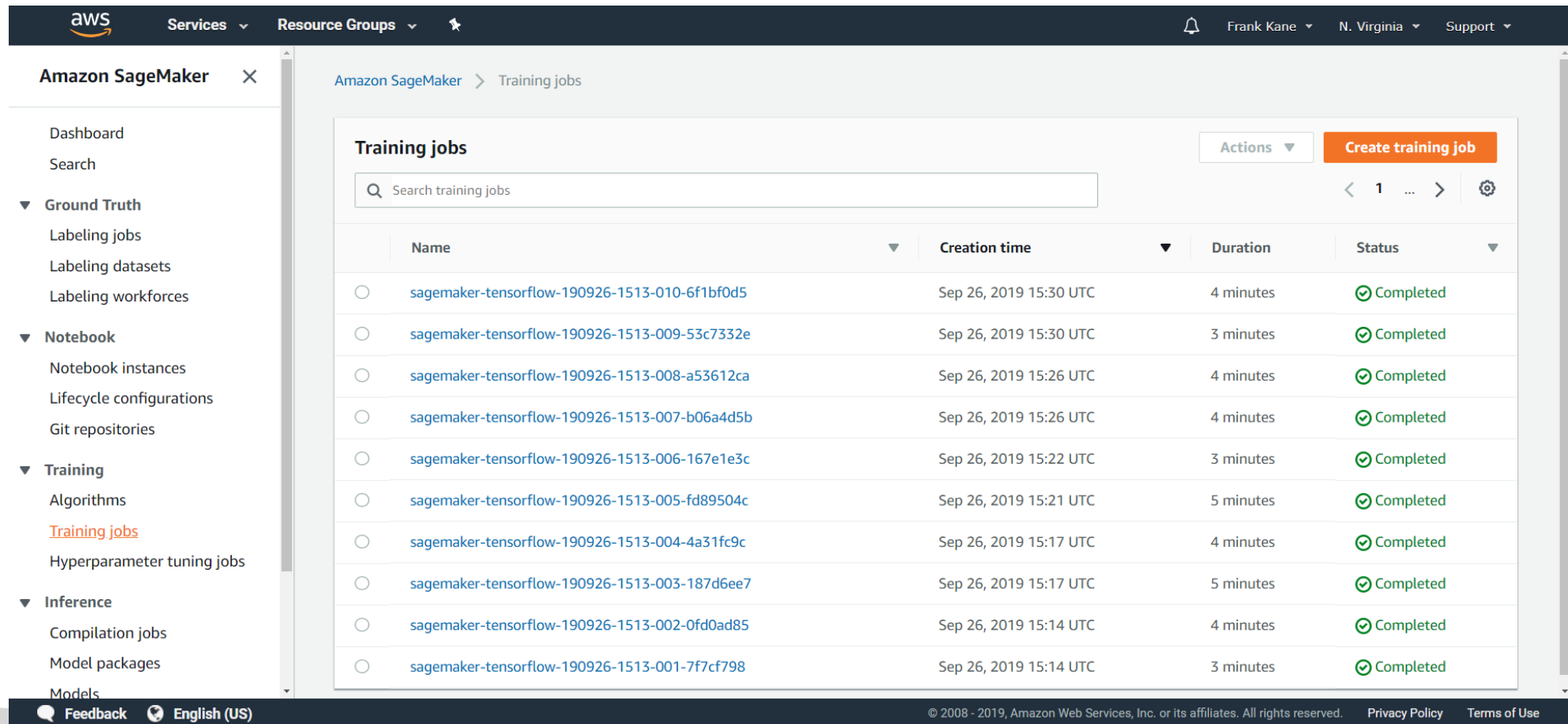
s3://sagemaker-us-east-1-159107795666/keras-mnist/training/training.npz
s3://sagemaker-us-east-1-159107795666/keras-mnist/validation/validation.npz
```

Below the code, a note states: "Note that sess.upload_data automatically creates an S3 bucket that meets the security criteria of starting with 'sagemaker-'."
- Test out our CNN training script locally on the notebook instance**: This section contains a code cell (In [4]) with the following code:

```
!pygmentize mnist-train-cnn.py
```

Below the code, a note states: "We'll test out running a single epoch, just to make sure the script works before we start spending money on P3 instances to train it further."

So can the SageMaker console

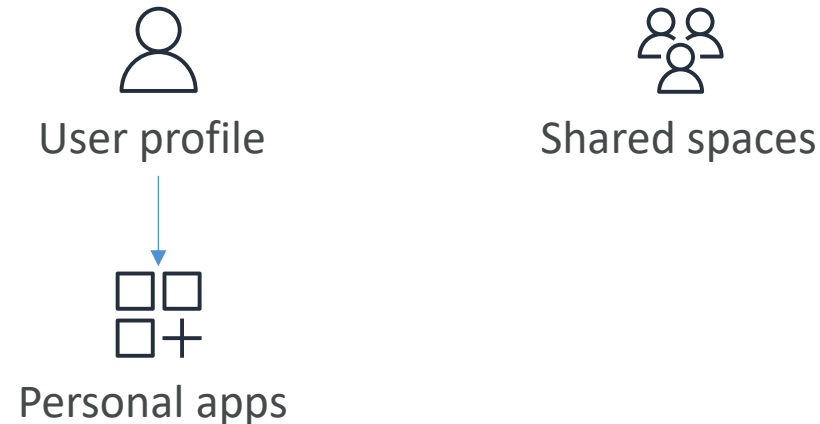
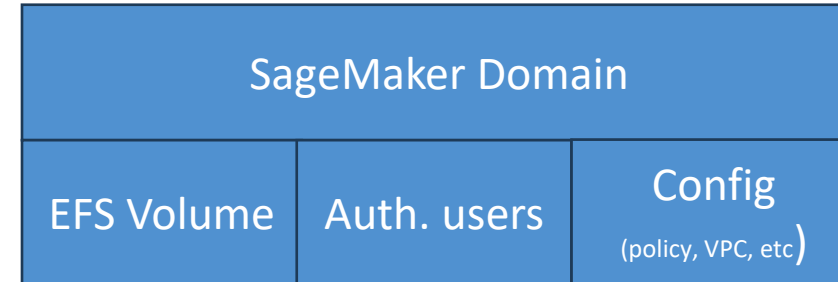


The screenshot displays the Amazon SageMaker console interface. On the left is a navigation sidebar with categories like Ground Truth, Notebook, Training, and Inference. The 'Training' section is expanded, showing 'Training jobs' as the selected option. The main content area, titled 'Training jobs', features a search bar and a table of job details. The table includes columns for Name, Creation time, Duration, and Status. All listed jobs are in a 'Completed' status. A 'Create training job' button is visible in the top right of the main area.

Name	Creation time	Duration	Status
sagemaker-tensorflow-190926-1513-010-6f1bf0d5	Sep 26, 2019 15:30 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-009-53c7332e	Sep 26, 2019 15:30 UTC	3 minutes	Completed
sagemaker-tensorflow-190926-1513-008-a53612ca	Sep 26, 2019 15:26 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-007-b06a4d5b	Sep 26, 2019 15:26 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-006-167e1e3c	Sep 26, 2019 15:22 UTC	3 minutes	Completed
sagemaker-tensorflow-190926-1513-005-fd89504c	Sep 26, 2019 15:21 UTC	5 minutes	Completed
sagemaker-tensorflow-190926-1513-004-4a31fc9c	Sep 26, 2019 15:17 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-003-187d6ee7	Sep 26, 2019 15:17 UTC	5 minutes	Completed
sagemaker-tensorflow-190926-1513-002-0fd0ad85	Sep 26, 2019 15:14 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-001-7f7cf798	Sep 26, 2019 15:14 UTC	3 minutes	Completed

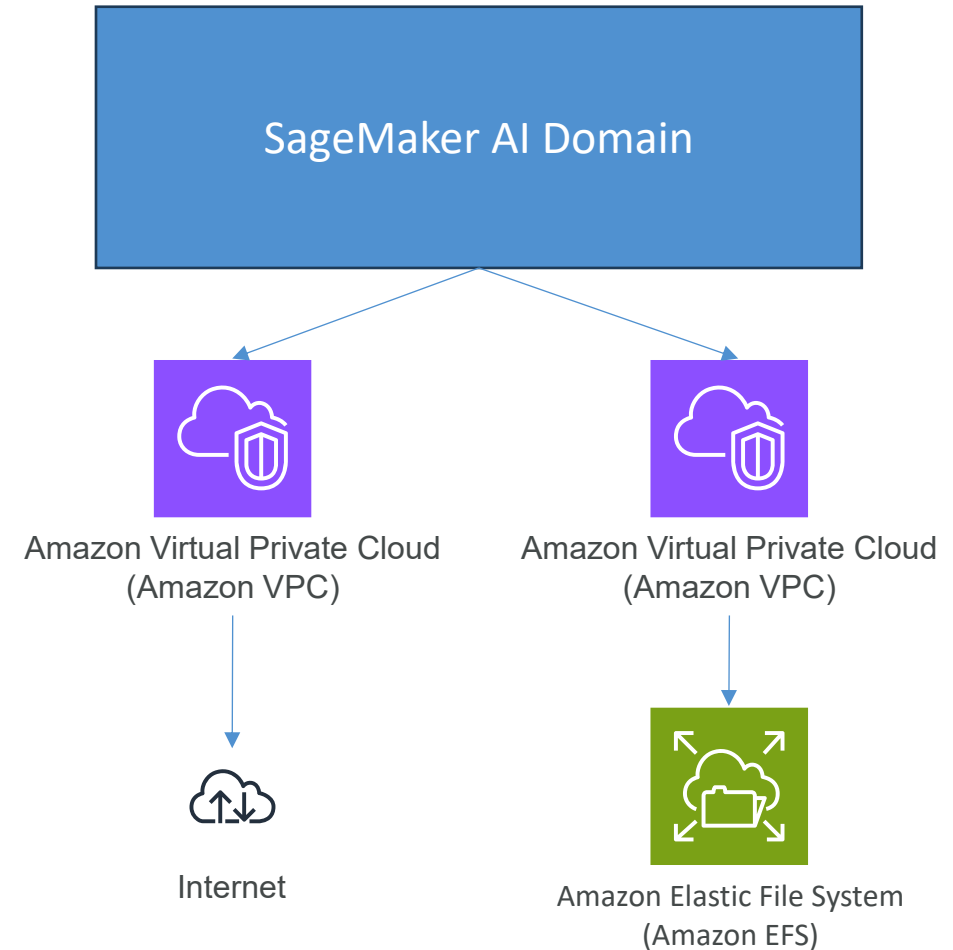
SageMaker AI Domains

- Domains organize users, apps, and resources
 - Share an EFS volume
- User profiles own personal apps
 - SageMaker Studio instances
 - Private EFS directory
 - Shared resources across other users
- Shared spaces
 - Shared EFS directory
 - Communal IDE app



VPC's for SageMaker AI Domains

- By default, a domain has two VPC's
 - One for Internet access
 - Managed by SageMaker AI
 - And your own
 - Encrypted traffic to your EFS volume
 - You specify the VPC, its subnets, and security groups
- You can change this
 - Send all traffic to your own VPC
 - Select “VPC Only” when creating the domain



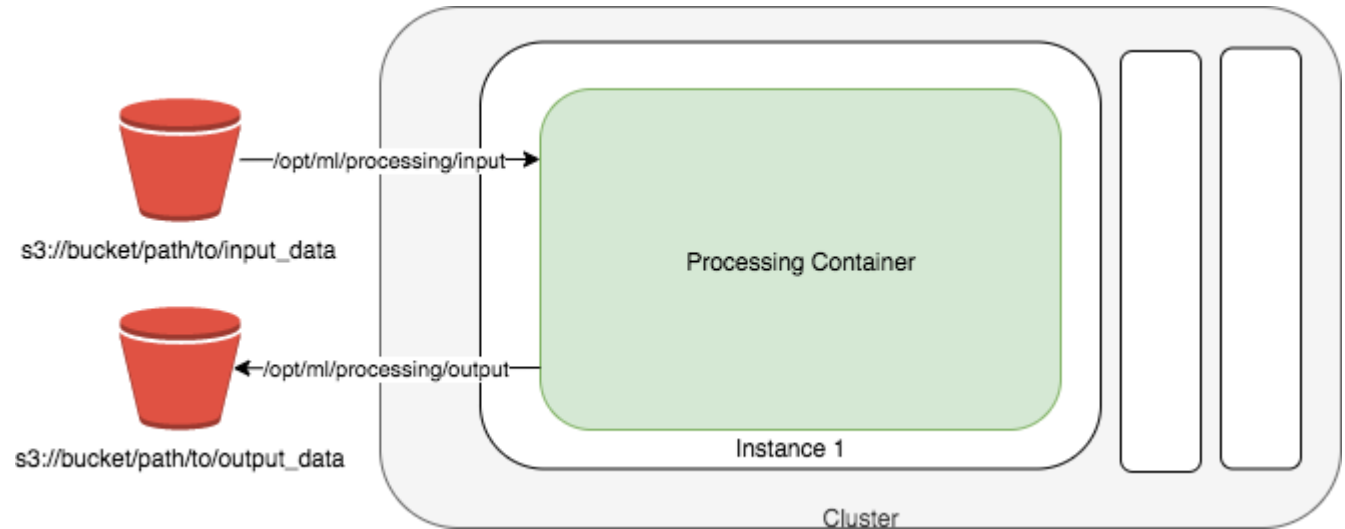
Data prep on SageMaker

- Data usually comes from S3
 - Ideal format varies with algorithm – often it is RecordIO / Protobuf
- Can also ingest from Athena, EMR, Redshift, and Amazon Keyspaces DB
- Apache Spark integrates with SageMaker
 - More on this later...
- Scikit_learn, numpy, pandas all at your disposal within a notebook



SageMaker Processing

- Processing jobs
 - Copy data from S3
 - Spin up a processing container
 - SageMaker built-in or user provided
 - Output processed data to S3



Training on SageMaker

- Create a training job
 - URL of S3 bucket with training data
 - ML compute resources
 - URL of S3 bucket for output
 - ECR path to training code
- Training options
 - Built-in training algorithms
 - Spark MLlib
 - Custom Python Tensorflow / MXNet code
 - PyTorch, Scikit-Learn, RLEstimator
 - XGBoost, Hugging Face, Chainer
 - Your own Docker image
 - Algorithm purchased from AWS marketplace



Deploying Trained Models

- Save your trained model to S3
- Can deploy two ways:
 - Persistent endpoint for making individual predictions on demand
 - SageMaker Batch Transform to get predictions for an entire dataset
- Lots of cool options
 - Inference Pipelines for more complex processing
 - SageMaker Neo for deploying to edge devices
 - Elastic Inference for accelerating deep learning models
 - Automatic scaling (increase # of endpoints as needed)
 - Shadow Testing evaluates new models against currently deployed model to catch errors



SageMaker Ground Truth

What is Ground Truth?

- Sometimes you don't have training data at all, and it needs to be generated by humans first.
- Example: training an image classification model. Somebody needs to tag a bunch of images with what they are images of before training a neural network
- Ground Truth manages humans who will label your data for training purposes

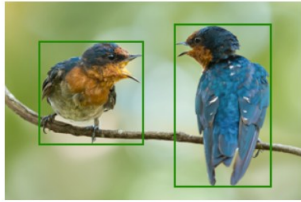
Select the task that a human worker will perform to label objects in your dataset.

☒ **Image classification**
Get workers to categorize images into specific classes. [Info](#)

☒ Basketball
☐ Soccer



☐ **Bounding box**
Get workers to draw bounding boxes around specified objects in your images. [Info](#)

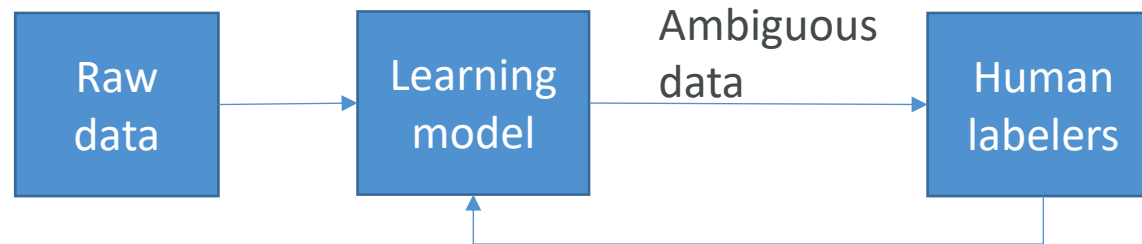


☐ **Semantic segmentation**
Get workers to draw pixel level labels around specific objects and segments in your images. [Info](#)



But it's more than that

- Ground Truth creates its own model as images are labeled by people
- As this model learns, only images the model isn't sure about are sent to human labelers
- This can reduce the cost of labeling jobs by 70%



Who are these human labelers?

- Mechanical Turk
- Your own internal team
- Professional labeling companies



Ground Truth Plus

- Turnkey solution
- “Our team of AWS Experts” manages the workflow and team of labelers
 - You fill out an intake form
 - They contact you and discuss pricing
- You track progress via the Ground Truth Plus Project Portal
- Get labeled data from S3 when done

Amazon SageMaker > Ground Truth Plus > Request a pilot

Request a pilot

We offer a free of cost pilot for customers who are looking to easily create high-quality training datasets without having to build labeling applications and manage the labeling workforce on their own.

Please fill out the project requirement form and our team will schedule a call with you to discuss and set up your data labeling project.

General information

Business email address

First name

Last name

Data type - *optional*

☐ Video

☐ Image

☐ 3D point clouds

☐ Document/text

☐ Audio

How do you label your data today? - *optional*

How many hours per week do you spend managing labeling operations? - *optional*

Labeling-input data format - *optional*
Example: .jpg, .jpeg, .png, .txt, .csv

Labeling-output data format - *optional*
Example: .csv, .txt

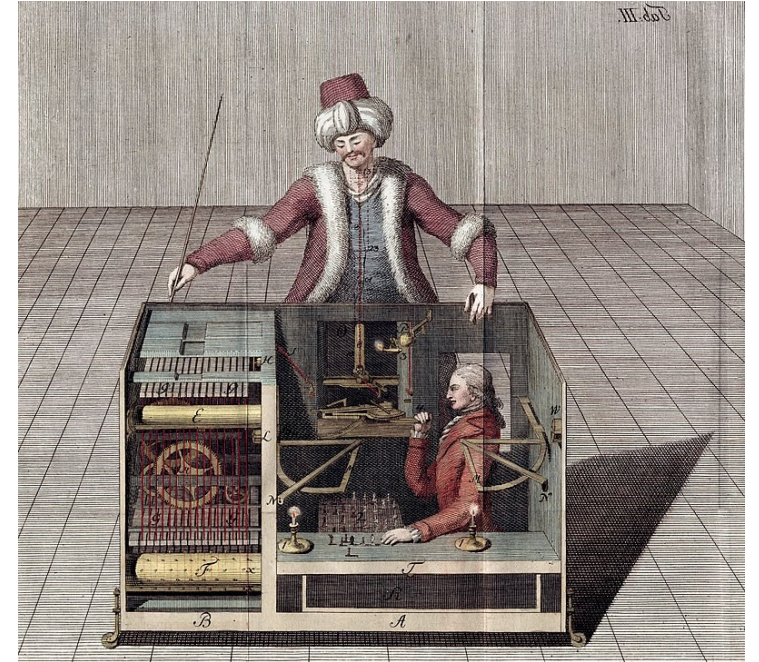
Cancel **Request a pilot**

Other ways to generate training labels

- Rekognition
 - AWS service for image recognition
 - Automatically classify images
- Comprehend
 - AWS service for text analysis and topic modeling
 - Automatically classify text by topics, sentiment
- Any pre-trained model or unsupervised technique that may be helpful

Amazon Mechanical Turk

- Crowdsourcing marketplace to perform simple human tasks
- Distributed virtual workforce
- Example:
 - You have a dataset of 10,000,000 images and you want to label these images
 - You distribute the task on Mechanical Turk and **humans** will tag those images
 - You set the reward per image (for example \$0.10 per image)
- Use cases: image classification, data collection, business processing
- **Integrates with Amazon A2I, SageMaker Ground Truth...**



Amazon Mechanical Turk

amazonmturk

Worker

HITs

Dashboard

Qualifications

Search All HITs

Q

Filter

All HITs

Your HITs Queue

HIT Groups (1-20 of 586)

+ Show Details

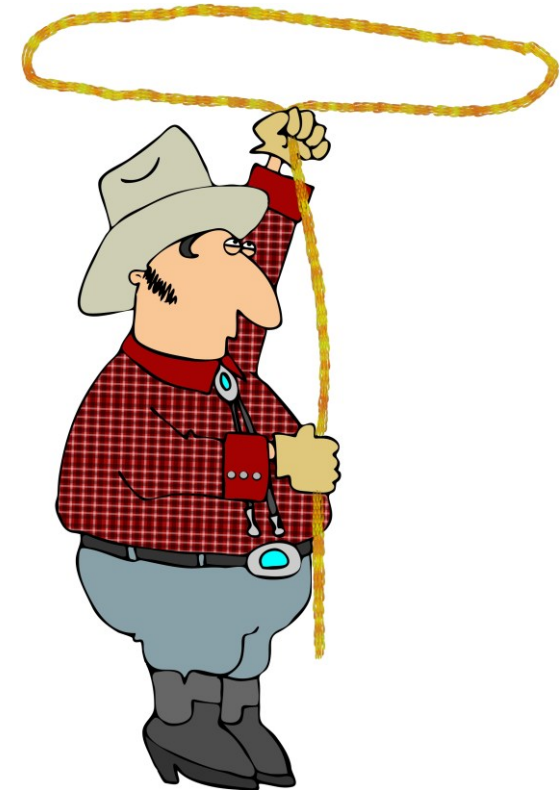
- Hide Details

Items Per Page: 20

Requester	Title	HITs	Reward	Created	Actions	
	Sentiment Annotation	13,210	\$0.01	1h ago	Preview	Accept & Work
	Transcribe up to 35 Seconds of Media to Text - Earn up to \$0.17 per HIT!!	11,237	\$0.05	21s ago	Preview	Qualify
	Market Research Survey	7,423	\$0.01	8m ago	Preview	Accept & Work
	Ask and answer questions about an image (V3)	5,428	\$0.22	11h ago	Preview	Qualify
	Collect Attorney Profile data from LinkedIn Website	4,430	\$0.05	5d ago	Preview	Qualify
	Quick survey	3,973	\$0.25	4h ago	Preview	Qualify
	Find the address for these rental listings44	2,926	\$3.50	1d ago	Preview	Qualify
	Object Segmentation in Image	2,267	\$0.50	2d ago	Preview	Accept & Work
	Reformatting Text	1,473	\$0.05	6d ago	Preview	Accept & Work
	Find and select a described person	1,229	\$0.05	2d ago	Preview	Accept & Work
	Find URLs for Hotels	946	\$0.50	2d ago	Preview	Qualify
	Point on heads/faces in images (Bonus for every HIT)	836	\$0.20	1h ago	Preview	Accept & Work

SageMaker Data Wrangler

- Visual interface (in SageMaker Studio) to prepare data for machine learning
- Import data
- Visualize data
- Transform data (300+ transformations to choose from)
 - Or integrate your own custom xforms with pandas, PySpark, PySpark SQL
- “Quick Model” to train your model with your data and measure its results

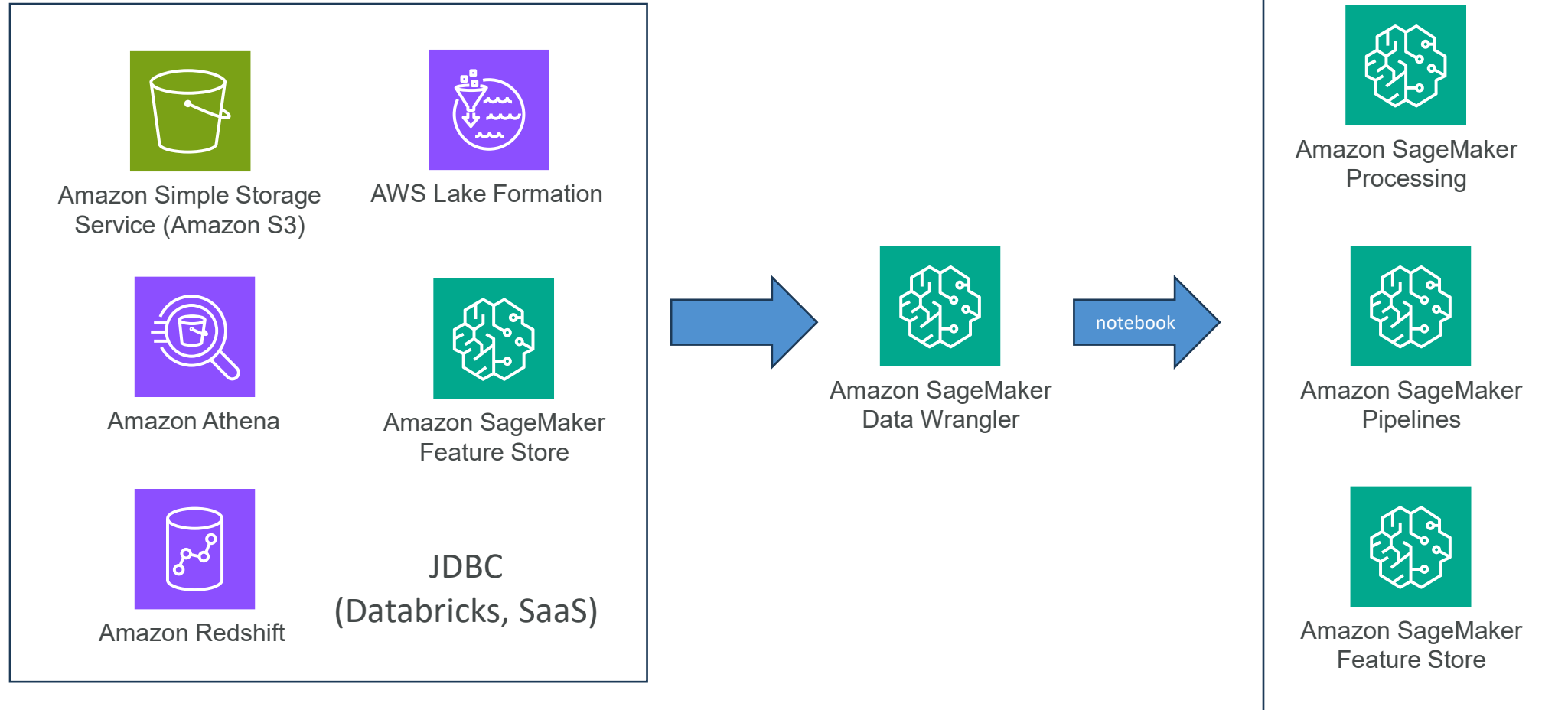


Data Wrangler: Notable Capabilities

- Transforming image data
 - Resize, enhance, corrupt images & more
- Balance data
 - Random oversampling, undersampling, SMOTE
- Impute missing data
- Handle outliers
- Dimensionality reduction (PCA)



Data Wrangler sources



Data Wrangler: Import Data

The screenshot shows the Amazon SageMaker Studio interface with the Data Wrangler tool open. The 'Import' tab is active, displaying a table of S3 objects for the 'titanic' dataset. The table lists one object: 'titanic-train.csv' with a size of 58.89KB and a last modified date of 2020-10-15 08:49:45+00:00. Below the table is a 'Preview' section showing the first five rows of the dataset. On the right, the 'DETAILS' panel shows the dataset name 'titanic-train.csv', the URI 's3://sagemaker-us-east-2-...', and file type 'csv'. Checkboxes for 'Add header to table' and 'Enable sampling' are both checked. An 'Import dataset' button is at the bottom of the details panel.

Amazon SageMaker Studio

File Edit View Run Kernel Git Tabs Settings Help

titanic.flow

Import Prepare Analyze Export

Data sources / S3 source / sagemaker-us-east-2-613904931467 / titanic

Import a dataset from S3

Use the following table to browse S3. Select a file to see import options. The following file formats are supported: CSV and Parquet.

Object Name	Size	Last Modified
titanic-train.csv	58.89KB	2020-10-15 08:49:45+00:00

Previous Displaying 1 - 1 Next

Preview

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
1	0	3	Braund, Mr. Owen Harris	male	22	1
2	1	1	Cumings, Mrs. John Bra...	female	38	1
3	1	3	Heikkinen, Miss. Laina	female	26	0
4	1	1	Futrelle, Mrs. Jacques H...	female	35	1
5	0	3	Allen, Mr. William Henry	male	35	0

DETAILS

Name

titanic-train.csv

Required

URI

s3://sagemaker-us-east-2-

File type

csv

Required

☒ Add header to table

☒ Enable sampling

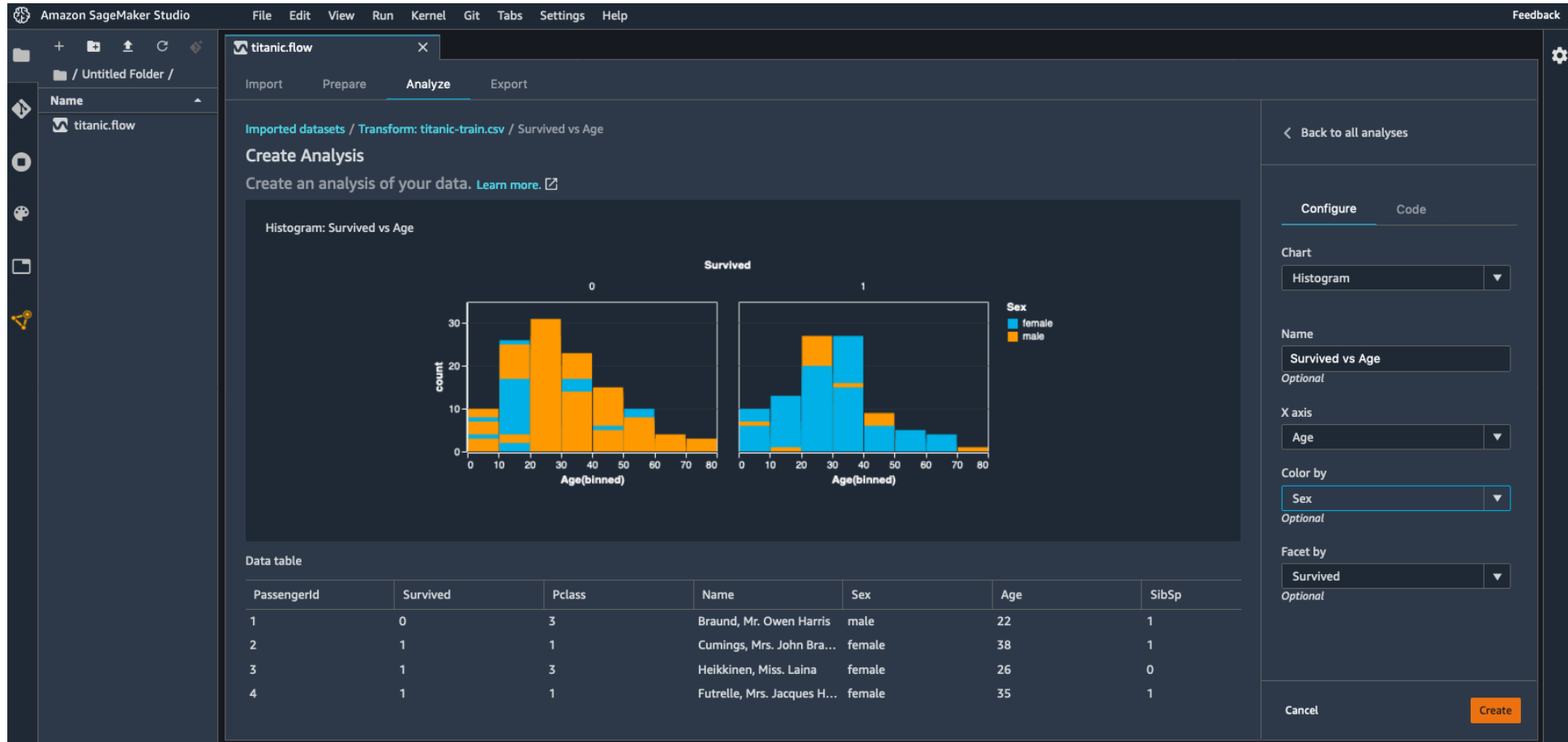
Import dataset

Data Wrangler: Preview Data

The screenshot displays the Amazon SageMaker Studio interface, specifically the Data Wrangler tool. The main window shows a data flow for 'titanic-train.csv' in the 'Prepare' stage. A table preview is visible, showing columns: Passengerid (long), Survived (long), Pclass (long), Name (string), Sex (string), and Age (long). The table contains 28 rows of data. On the right, the 'CONFIGURE TYPES' panel allows users to select data types for each column. The selected types are: Passengerid (Long), Survived (Long), Pclass (Long), Name (String), Sex (String), Age (Long), SibSp (Long), Parch (Long), Ticket (String), Fare (Float), Cabin (String), and Embarked (String). Buttons for 'Clear', 'Preview', and 'Apply' are at the bottom of the configuration panel.

Passengerid (long)	Survived (long)	Pclass (long)	Name (string)	Sex (string)	Age (long)
7	0	1	McCarthy, Mr. Timothy J	male	54
8	0	3	Palsson, Master. Gosta ...	male	2
9	1	3	Johnson, Mrs. Oscar W (...)	female	27
10	1	2	Nasser, Mrs. Nicholas (A...	female	14
11	1	3	Sandstrom, Miss. Margu...	female	4
12	1	1	Bonnell, Miss. Elizabeth	female	58
13	0	3	Saunderscock, Mr. Willia...	male	20
14	0	3	Andersson, Mr. Anders J...	male	39
15	0	3	Vestrom, Miss. Hulda A...	female	14
16	1	2	Hewlett, Mrs. (Mary D K...	female	55
17	0	3	Rice, Master. Eugene	male	2
18	1	2	Williams, Mr. Charles Eu...	male	
19	0	3	Vander Planke, Mrs. Juli...	female	31
20	1	3	Masselmani, Mrs. Fatima	female	
21	0	2	Fynney, Mr. Joseph J	male	35
22	1	2	Beesley, Mr. Lawrence	male	34
23	1	3	McGowan, Miss. Anna "...	female	15
24	1	1	Sloper, Mr. William Tho...	male	28
25	0	3	Palsson, Miss. Torborg ...	female	8
26	1	3	Asplund, Mrs. Carl Osca...	female	38
27	0	3	Emir, Mr. Farred Chehab	male	
28	0	1	Fortune, Mr. Charles Ale...	male	19

Data Wrangler: Visualize Data



Data Wrangler: Transform Data

The screenshot shows the Amazon SageMaker Studio interface with the Data Wrangler tool open. The 'Prepare' tab is active, and the 'Encode categorical' transform is being configured for the 'Pclass' column. The transform is set to 'One-hot encode' with 'Keep' as the invalid handling strategy. The output format is 'Flattened'.

Transform: titanic-train.csv

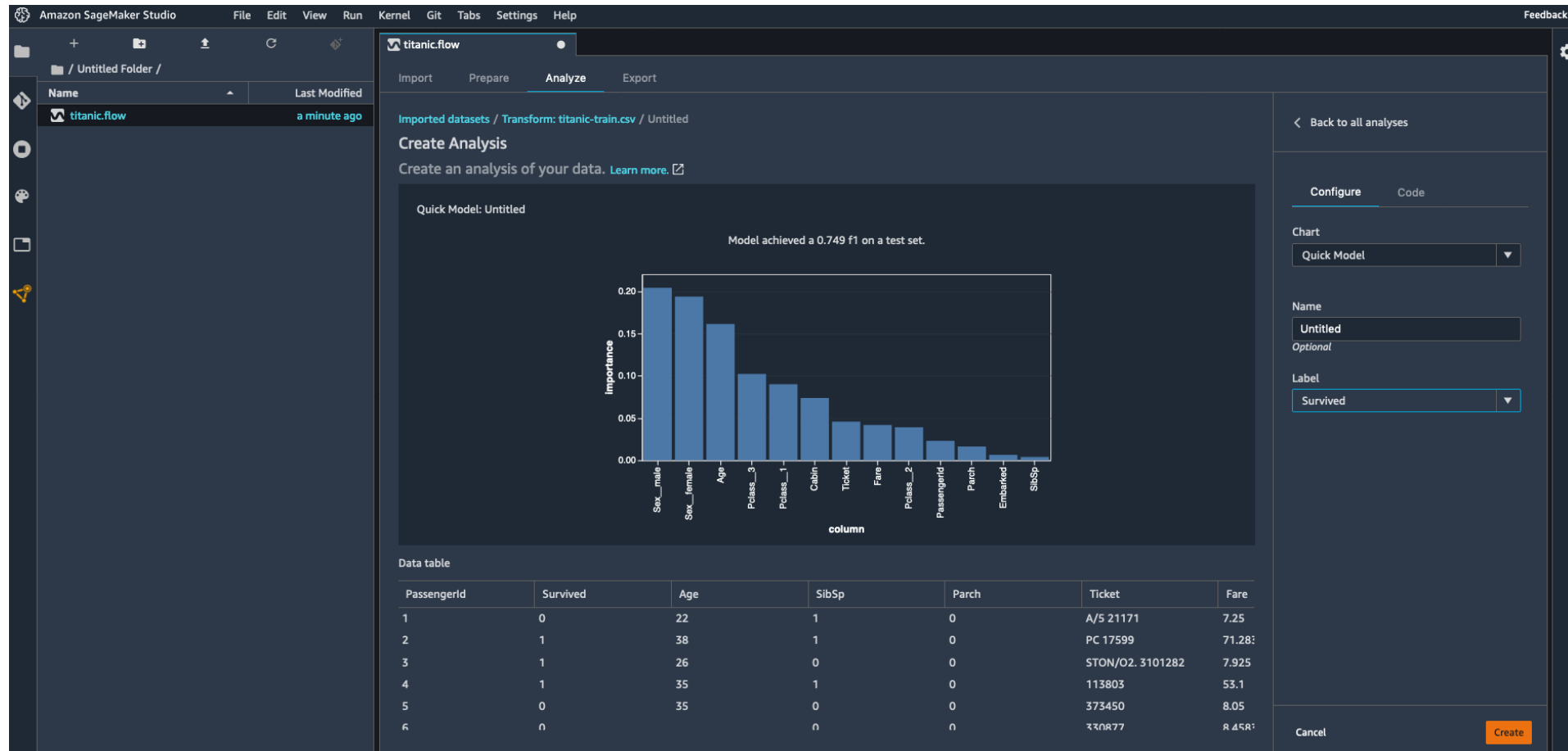
t)	Cabin (string)	Embarked (string)	Pclass_3 (float)	Pclass_1 (float)	Pclass_2 (float)
		S	1	0	0
	C85	C	0	1	0
		S	1	0	0
	C123	S	0	1	0
		S	1	0	0
		Q	1	0	0
	E46	S	0	1	0
		S	1	0	0
		C	0	0	1
	G6	S	1	0	0
	C103	S	0	1	0
		S	1	0	0
		S	1	0	0
		S	1	0	0
		S	0	0	1
		Q	1	0	0
		S	0	0	1
		S	1	0	0

Configuration:

- Input column: Pclass
- Output column name: Pclass_
- Transform: One-hot encode
- Invalid handling strategy: Keep
- Drop last category: Select...
- Is input ordinal encoded?: Select...
- Output format: Flattened

Buttons: Clear, Preview, Add

Data Wrangler: Quick Model



Data Wrangler: Export Data Flow

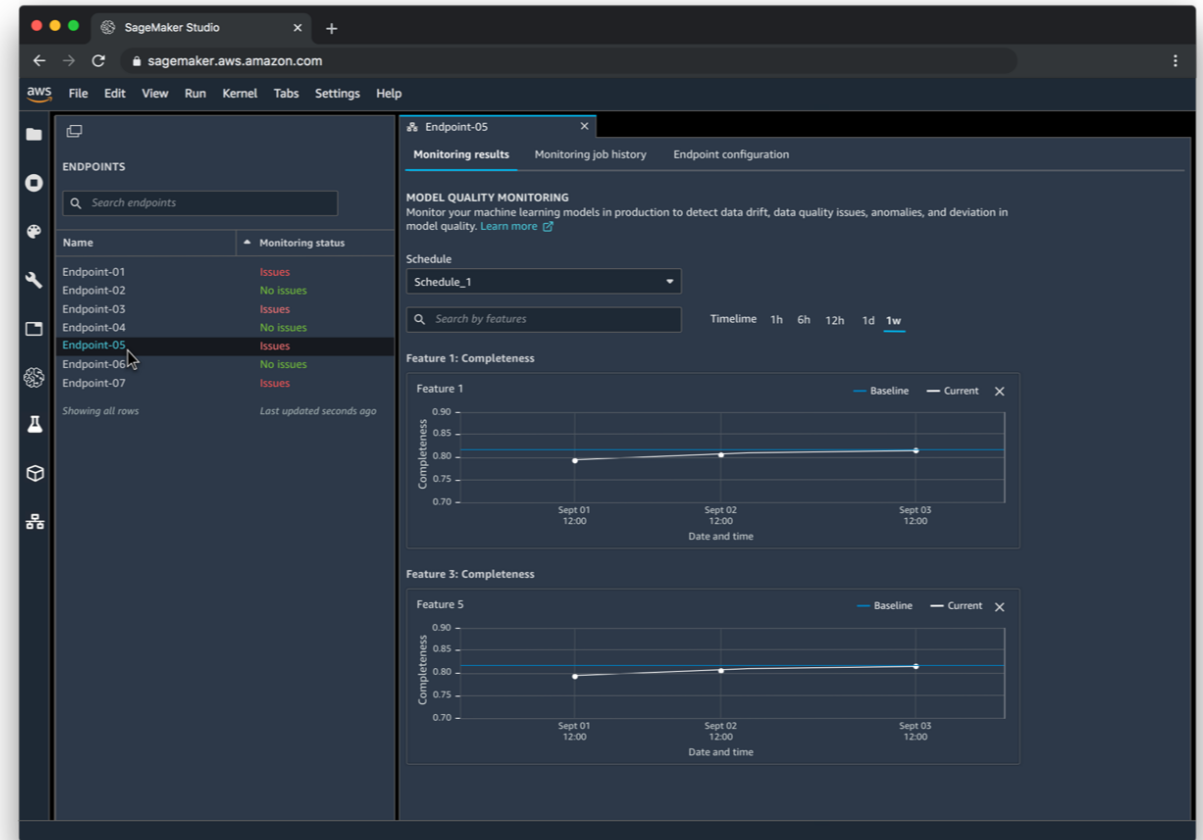
The screenshot displays the Amazon SageMaker Studio interface, specifically the Data Wrangler tool. The top menu bar includes 'File', 'Edit', 'View', 'Run', 'Kernel', 'Git', 'Tabs', 'Settings', and 'Help'. The left sidebar shows a file explorer with a folder named 'Untitled Folder' containing a file 'titanic.flow' modified 'a minute ago'. The main workspace is titled 'titanic.flow' and has tabs for 'Import', 'Prepare', 'Analyze', and 'Export'. The 'Export' tab is active, showing the 'Export data flow' section. Below the title, it says 'Select the steps in the data flow that you want to export and then select an export option.' The data flow diagram consists of three steps: 'Source - sampled' (S3: titanic-train.csv), 'Data types' (Transform: titanic-train.csv), and 'Steps (5)' (a code icon). The 'Steps (5)' step is highlighted with a blue box. On the right, there is a panel with four export options: 'Processing Job' (Export a Jupyter Notebook that launches a processing job.), 'Pipeline' (Export a Jupyter Notebook that creates a Pipeline with your data flow.), 'Python Code' (Export your data flow to python code.), and 'Feature Store' (Export a Jupyter Notebook that creates a Feature Store feature group and adds features to an offline or online feature store.). An 'Export step' button is located at the top right of this panel.

Data Wrangler Troubleshooting

- Make sure your Studio user has appropriate IAM roles
- Make sure permissions on your data sources allow Data Wrangler access
 - Add AmazonSageMakerFullAccess policy
- EC2 instance limit
 - If you get “The following instance type is not available...” errors
 - May need to request a quota increase
 - Service Quotas / Amazon SageMaker / Studio KernelGateway Apps running on ml.m5.4xlarge instance

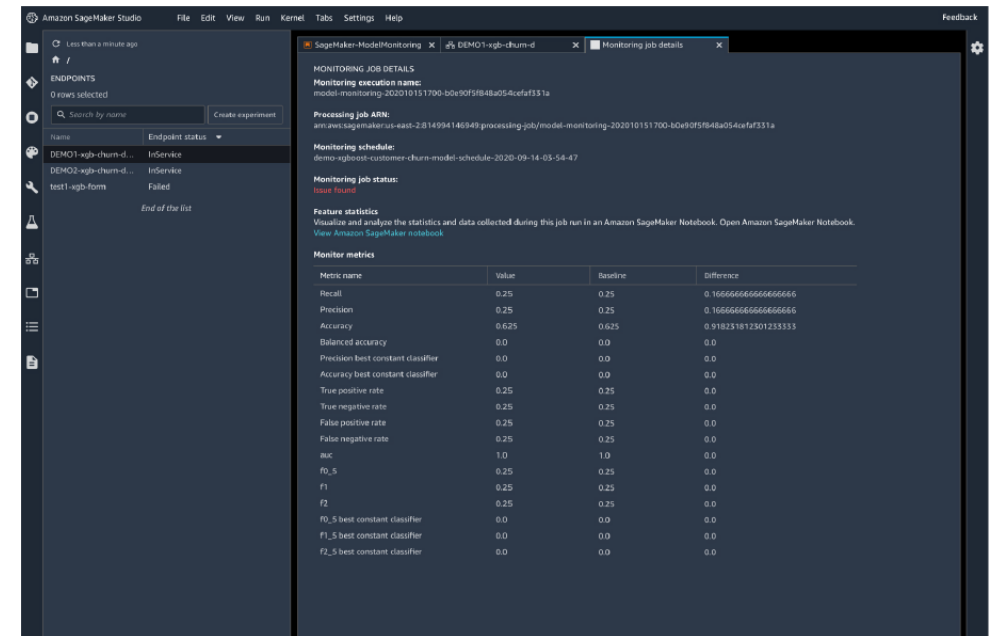
SageMaker Model Monitor

- Get alerts on quality deviations on your deployed models (via CloudWatch)
- Visualize data drift
 - Example: loan model starts giving people more credit due to drifting or missing input features
- Detect anomalies & outliers
- Detect new features



SageMaker Model Monitor + Clarify

- Integrates with SageMaker Clarify
 - SageMaker Clarify detects potential bias
 - i.e., imbalances across different groups / ages / income brackets
 - With ModelMonitor, you can monitor for bias and be alerted to new potential bias via CloudWatch
 - SageMaker Clarify also helps explain model behavior
 - Understand which features contribute the most to your predictions



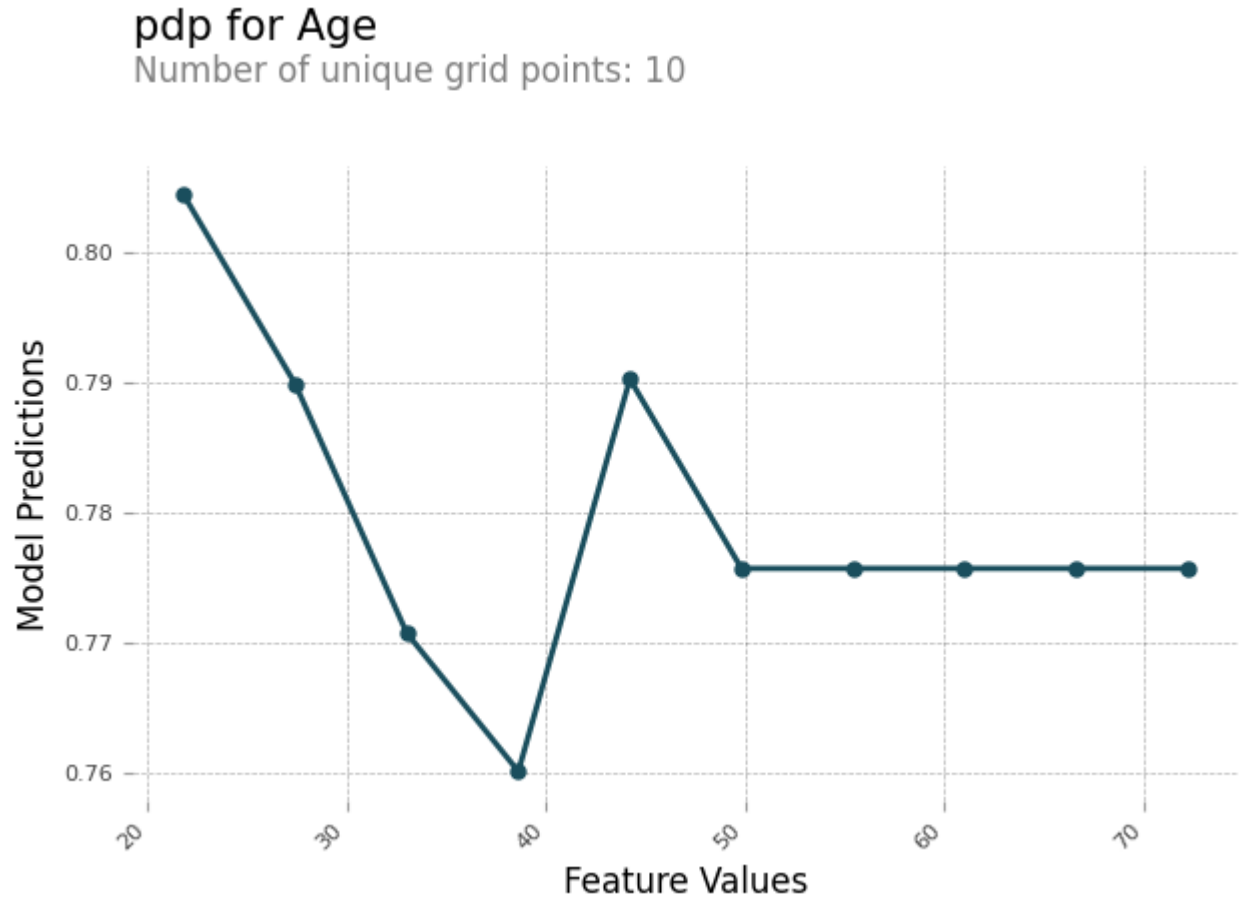
Pre-training Bias Metrics in Clarify

- Class Imbalance (CI)
 - One facet (demographic group) has fewer training values than another
- Difference in Proportions of Labels (DPL)
 - Imbalance of positive outcomes between facet values
- Kullback-Leibler Divergence (KL), Jensen-Shannon Divergence(JS)
 - How much outcome distributions of facets diverge
- Lp-norm (LP)
 - P-norm difference between distributions of outcomes from facets
- Total Variation Distance (TVD)
 - L1-norm difference between distributions of outcomes from facets
- Kolmogorov-Smirnov (KS)
 - Maximum divergence between outcomes in distributions from facets
- Conditional Demographic Disparity (CDD)
 - Disparity of outcomes between facets as a whole, and by subgroups



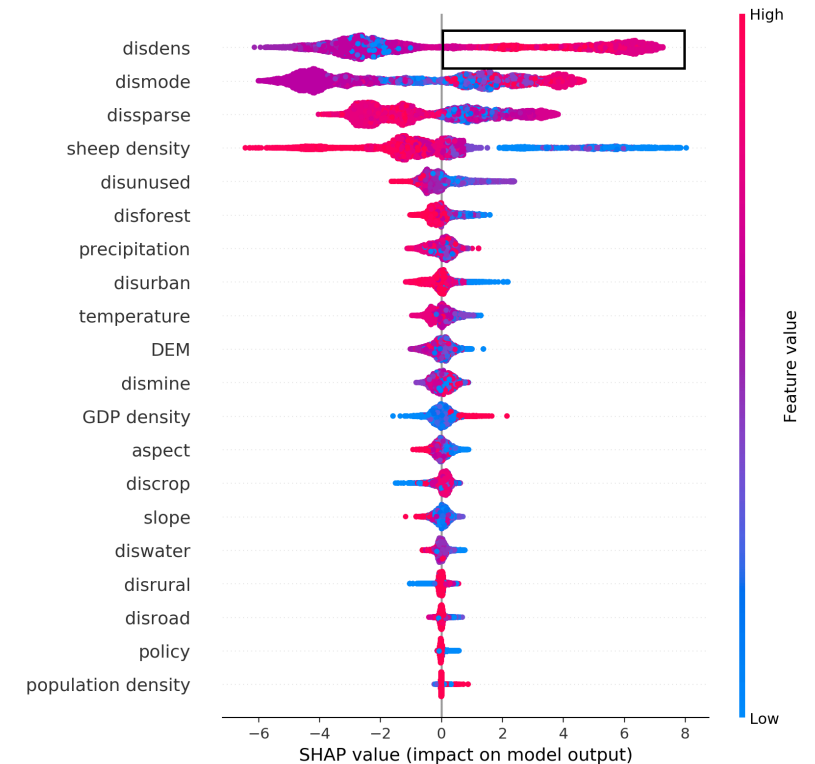
Partial Dependence Plots (PDPs)

- Shows dependence of predicted target response on a set of input features
- Plots can show you how feature values influence the predictions
 - In this example, higher values result in the same predictions
- You can also get back data distributions for each bucket of values



Shapley Values

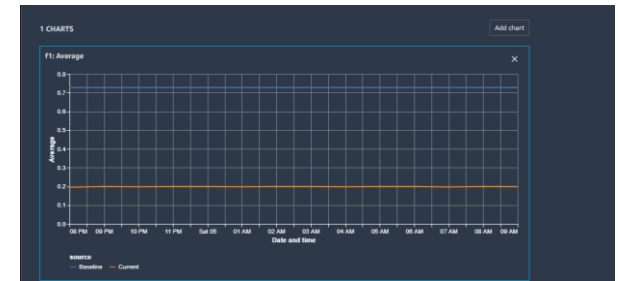
- Shapley values are the algorithm used to determine the contribution of each feature toward a model's predictions
 - Originated in game theory, adapted to ML
 - Basically measures the impact of dropping individual features
 - Gets complicated with lots of features
 - SageMaker Clarify uses Shapley Additive exPlanations (SHAP) as an approximation technique
- Asymmetric Shapley Values:
 - For Time Series
 - The algorithm used to determine the contribution of input features at each time step toward forecasted predictions



Batunacun, Wieland, R., Lakes, T., and Nendel, C.: Using Shapley additive explanations to interpret extreme gradient boosting predictions of grassland degradation in Xilingol, China, Geosci. Model Dev., 14, 1493–1510, <https://doi.org/10.5194/gmd-14-1493-2021>, 2021.

SageMaker Model Monitor

- Data is stored in S3 and secured
- Monitoring jobs are scheduled via a Monitoring Schedule
- Metrics are emitted to CloudWatch
 - CloudWatch notifications can be used to trigger alarms
 - You'd then take corrective action (retrain the model, audit the data)
- Integrates with Tensorboard, QuickSight, Tableau
- Or just visualize within SageMaker Studio



SageMaker Model Monitor

- Monitoring Types:
 - Drift in data quality
 - Relative to a baseline you create
 - “Quality” is just statistical properties of the features
 - Drift in model quality (accuracy, etc)
 - Works the same way with a model quality baseline
 - Can integrate with Ground Truth labels
 - Bias drift
 - Feature attribution drift
 - Based on Normalized Discounted Cumulative Gain (NDCG) score
 - This compares feature ranking of training vs. live data

AWS Managed AI Services

Why AWS AI Managed Services?

- **AWS AI Services are pre-trained ML services for your use case**
- **Responsiveness and Availability**
- **Redundancy and Regional Coverage:** deployed across multiple Availability Zones and AWS regions
- **Performance:** specialized CPU and GPUs for specific use-cases for cost saving
- **Token-based pricing:** pay for what you use
- **Provisioned throughput:** for predictable workloads, cost savings and predictable performance



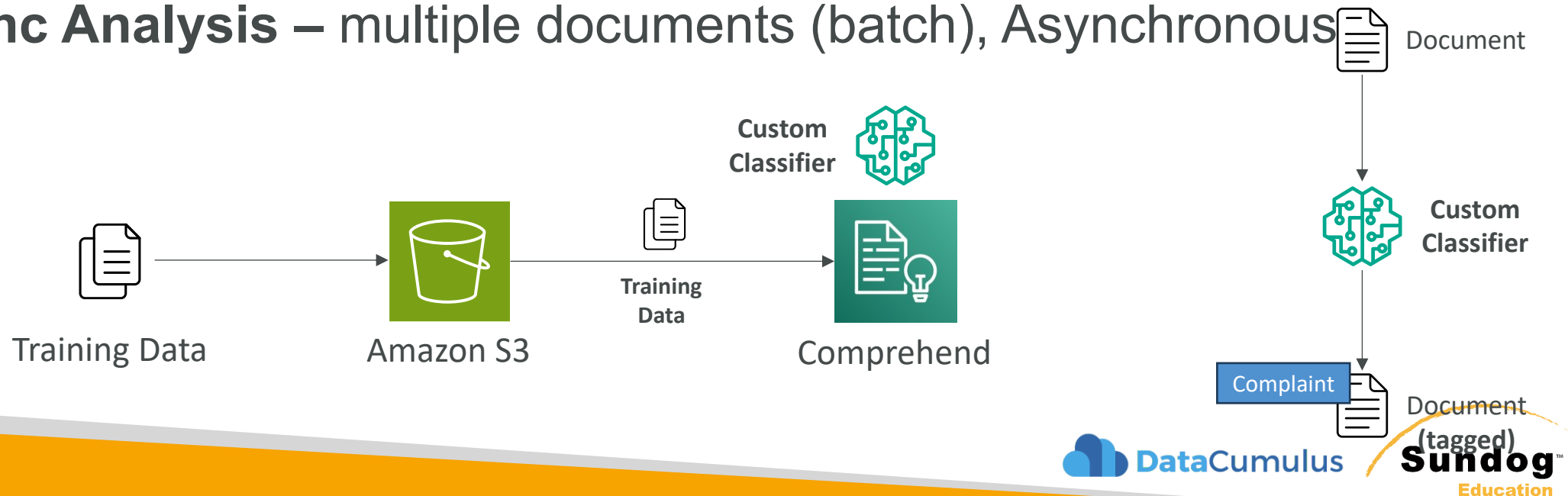
Amazon Comprehend



- For **Natural Language Processing – NLP**
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
 - Language of the text
 - Extracts key phrases, places, people, brands, or events
 - Understands how positive or negative the text is
 - Analyzes text using tokenization and parts of speech
 - Automatically organizes a collection of text files by topic
- Sample use cases:
 - analyze customer interactions (emails) to find what leads to a positive or negative experience
 - Create and groups articles by topics that Comprehend will uncover

Comprehend – Custom Classification

- Organize documents into categories (classes) that you define
- Example: categorize customer emails so that you can provide guidance based on the type of the customer request
- Supports different document types (text, PDF, Word, images...)
- **Real-time Analysis** – single document, synchronous
- **Async Analysis** – multiple documents (batch), Asynchronous



Named Entity Recognition (NER)

- **NER** – Extracts predefined, general-purpose entities like people, places, organizations, dates, and other standard categories, **from text**

Hello Zhang Wei, I am John. Your AnyCompany Financial Services, LLC credit card account 1111-0000-1111-0008 has a minimum payment of \$24.53 that is due by July 31st. Based on your autopay settings, we will withdraw your payment on the due date from your bank account number XXXXXX1111 with the routing number XXXXX0000.

Customer feedback for Sunshine Spa, 123 Main St, Anywhere. Send comments to Alice at sunspa@mail.com.

I enjoyed visiting the spa. It was also very expensive. The amenities were ok but the service made the spa a great experience.

Entity 10: 123 Main St ✕

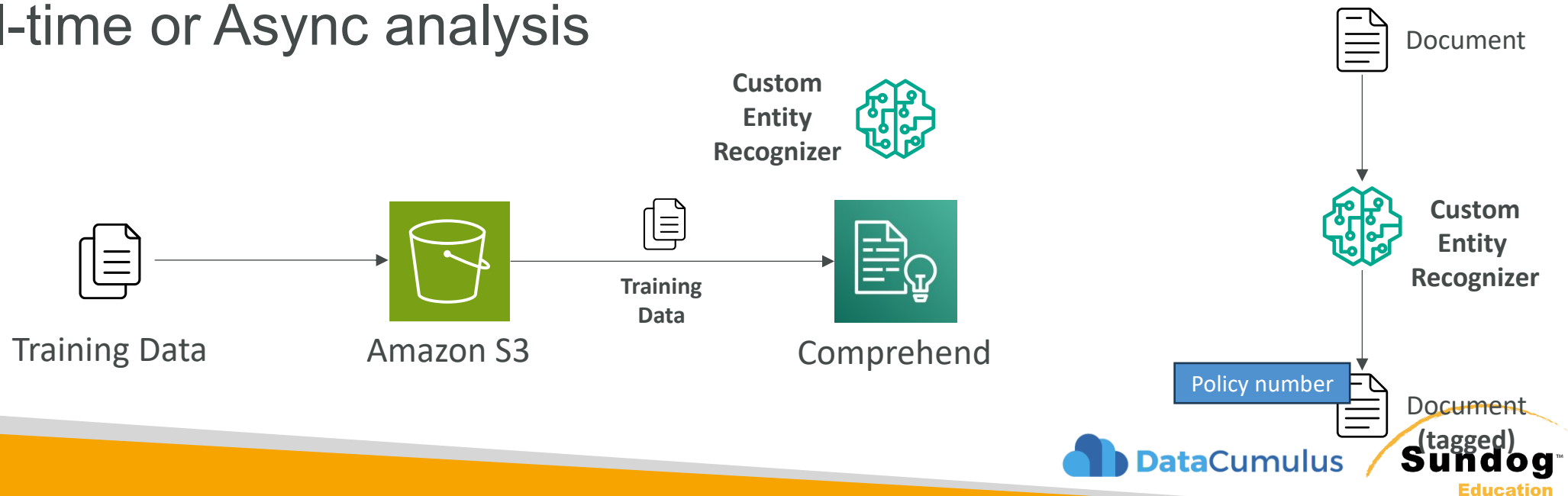
Entity: LOCATION

Confidence: 0.98

Entity ▾	Type ▾
Zhang Wei	Person
John	Person
AnyCompany Financial Services, LLC	Organization
1111-0000-1111-0008	Other
\$24.53	Quantity
July 31st	Date
XXXXXX1111	Other
XXXXX0000	Other
Sunshine Spa	Organization
123 Main St	Location

Comprehend – Custom Entity Recognition

- Analyze text for specific terms and noun-based phrases
- Extract terms like policy numbers, or phrases that imply a customer escalation, anything specific to your business
- Train the model with custom data such as a list of the entities and documents that contain them
- Real-time or Async analysis



Comprehend – Custom Models

- You can create custom models for entity recognition or document classification
 - Trained on your own data
- Comprehend manages your model versioning
- Custom models may be copied between AWS accounts
 - Attach IAM policy to a model version, authorizing the other account
 - Other account then imports the model
 - Must be in same region
 - Need its ARN, region, and optional KMS key
 - Can be done from the Comprehend console

Amazon Translate



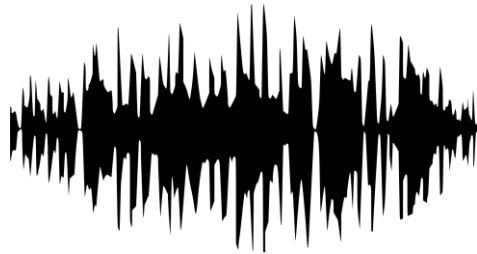
- Natural and accurate **language translation**
- Amazon Translate allows you to **localize content** - such as websites and applications - for **international users**, and to easily translate large volumes of text efficiently.

Source language		Target language
Auto (auto) ▼	↔	French (fr) ▼
Hi my name is Stéphane		Bonjour, je m'appelle Stéphane.
		Portuguese (pt) ▼
		Oi, meu nome é Stéphane.
		Hindi (hi) ▼
		हाय मेरा नाम स्टीफन है

Amazon Transcribe



- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition (ASR)** to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification** for multi-lingual audio
- Use cases:
 - transcribe customer service calls
 - automate closed captioning and subtitling
 - generate metadata for media assets to create a fully searchable archive



*"Hello my name is Stéphane.
I hope you're enjoying the course!"*

Transcribe – Toxicity Detection

- ML-powered, voice-based toxicity detection capability
- Leverages speech cues: tone and pitch, and text-based cues
- Toxicity categories: sexual harassment, hate speech, threat, abuse, profanity, insult, and graphic....

Transcription preview
You can see the first 5,000 characters of the transcription text below. To download the full text, choose Download full transcript.

Download ▾

Text Audio identification Subtitles **Toxicity**

Toxicity score -- 0.0 to 0.4 0.4 to 0.8 0.8 to 1.0 Info
Higher score means higher toxicity.

Hide filters

Filters Info Reset
Filter out toxic content by increasing threshold values below.

Toxicity Score 0 1
Profanity 0 1
Hate speech 0 1
Sexual 0 1
Insults 0 1

Transcription text preview (redacted):
... You are a waste of a human being. You've done ABSOLUTELY NOTHING that's important. ... You know, you're not even human. You don't deserve the word human because you are that much of a ... disgrace to people all around the globe, worldwide. It's a ... Are you proud of having accomplished absolutely nothing in your life? You have only done bad things in this world. You do not deserve to live, y ... t. ...

Toxicity Categories

Profanity: Speech that contains words, phrases, or acronyms that are impolite, vulgar, or offensive.

Hate speech: Speech that criticizes, insults, denounces, or dehumanizes a person or group on the basis of an identity (such as race, ethnicity, gender, religion, sexual orientation, ability, and national origin).

Sexual: Speech that indicates sexual interest, activity, or arousal using direct or indirect references to body parts, physical traits, or sex.

Insults: Speech that includes demeaning, humiliating, mocking, insulting, or belittling language. This type of language is also labeled as bullying.

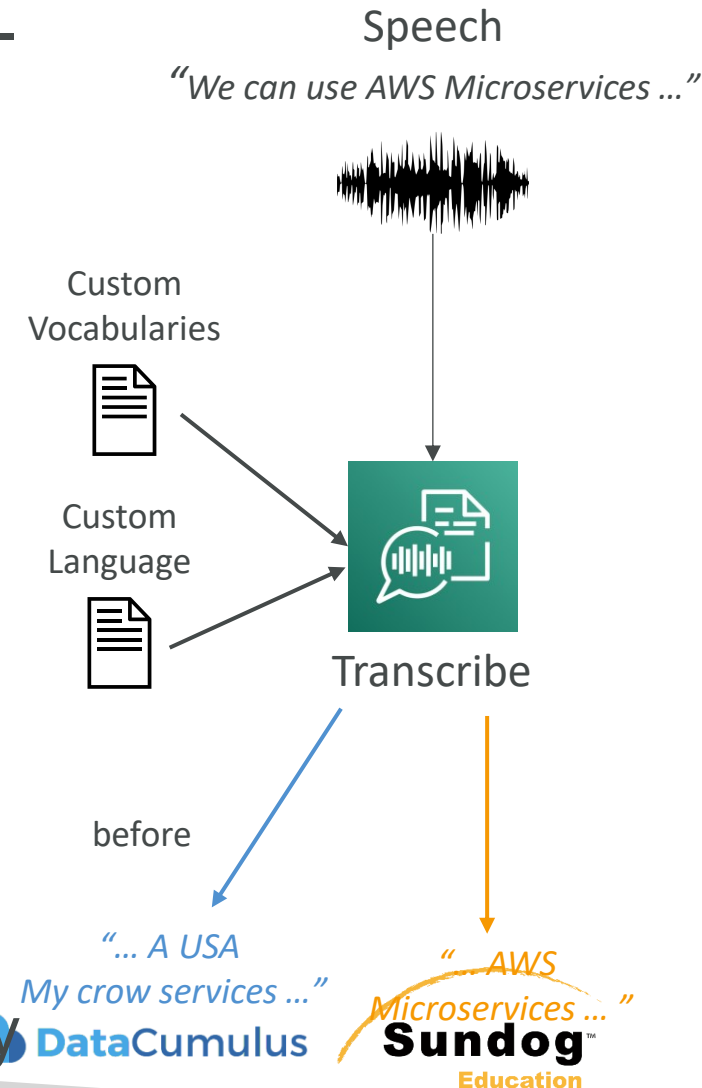
Violence or threat: Speech that includes threats seeking to inflict pain, injury, or hostility toward a person or group.

Graphic: Speech that uses visually descriptive and unpleasantly vivid imagery. This type of language is often intentionally verbose to amplify a recipient's discomfort.

Harassment or abusive: Speech intended to affect the psychological well-being of the recipient, including demeaning and objectifying terms.

Amazon Transcribe – Improving Accuracy

- Allows Transcribe to capture domain-specific or non-standard terms (e.g., technical words, acronyms, jargon...)
- **Custom Vocabularies (for words)**
 - Add specific words, phrases, domain-specific terms
 - Good for brand names, acronyms...
 - Increase recognition of a new word by providing hints (such as pronunciation..)
- **Custom Language Models (for context)**
 - Train Transcribe model on your own domain-specific text data
 - Good for transcribing large volumes of domain-specific speech
 - Learn the context associated with a given word

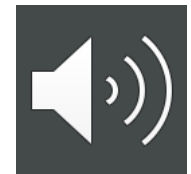


Amazon Polly



- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk

*Hi! My name is Stéphane
and this is a demo of Amazon Polly*



Polly – Advanced Features

- **Lexicons**

- Define how to read certain specific pieces of text
- AWS => “Amazon Web Services”
- W3C => “World Wide Web Consortium”

- **SSML - Speech Synthesis Markup Language**

- Markup for your text to indicate how to pronounce it
- Example: “Hello, <break> how are you?”

- **Voice engine:** generative, long-form, neural, standard...

- **Speech mark:**

- Encode where a sentence/word starts or ends in the audio
- Helpful for lip-syncing or highlight words as they’re spoken

Action	SSML tag
Adding a pause	<break>
Emphasizing words	<emphasis>
Specifying another language for specific words	<lang>
Placing a custom tag in your text	<mark>
Adding a pause between paragraphs	<p>
Using phonetic pronunciation	<phoneme>
Controlling volume, speaking rate, and pitch	<prosody>
Setting a maximum duration for synthesized speech	<prosody amazon:max-duration>
Adding a pause between sentences	<s>
Controlling how special types of words are spoken	<say-as>
Identifying SSML-enhanced text	<speak>
Pronouncing acronyms and abbreviations	<sub>
Improving pronunciation by specifying parts of speech	<w>
Adding the sound of breathing	<amazon:auto-breaths>
Newscaster speaking style	<amazon:domain name="news">
Adding dynamic range compression	<amazon:effect name="drc">
Speaking softly	<amazon:effect



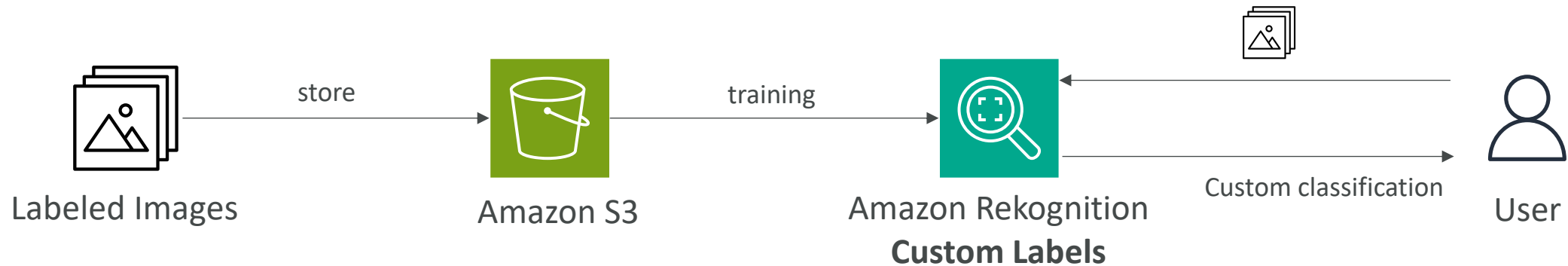
Amazon Rekognition



- Find **objects, people, text, scenes** in **images and videos** using ML
- **Facial analysis** and **facial search** to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities
- Use cases:
 - Labeling
 - Content Moderation
 - Text Detection
 - Face Detection and Analysis (gender, age range, emotions...)
 - Face Search and Verification
 - Celebrity Recognition
 - **Pathing** (ex: for sports game analysis)

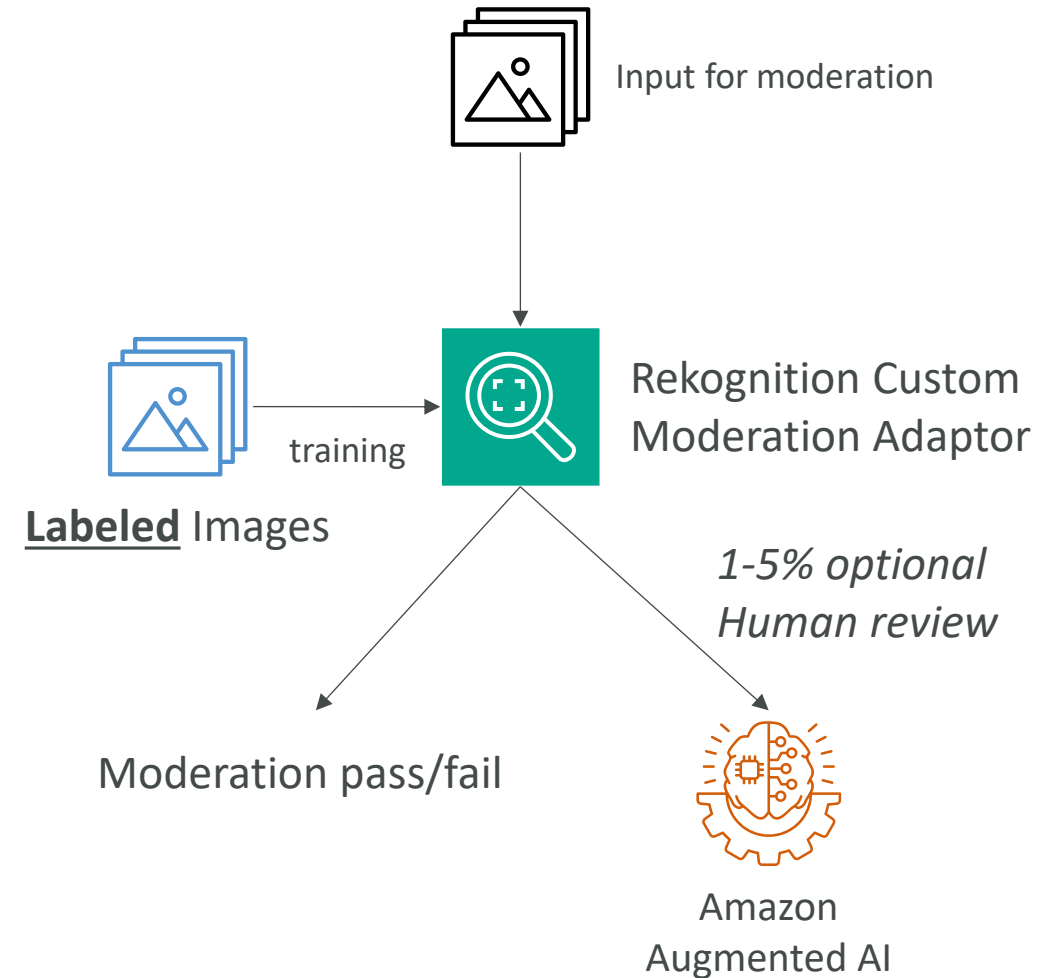
Amazon Rekognition – Custom Labels

- Examples: find your logo in social media posts, identify your products on stores shelves (National Football League – NFL – uses it to find their logo in pictures)
- **Label** your training images and upload them to Amazon Rekognition
- Only needs a few hundred images or less
- Amazon Rekognition creates a custom model on your images set
- New subsequent images will be categorized the custom way you have defined

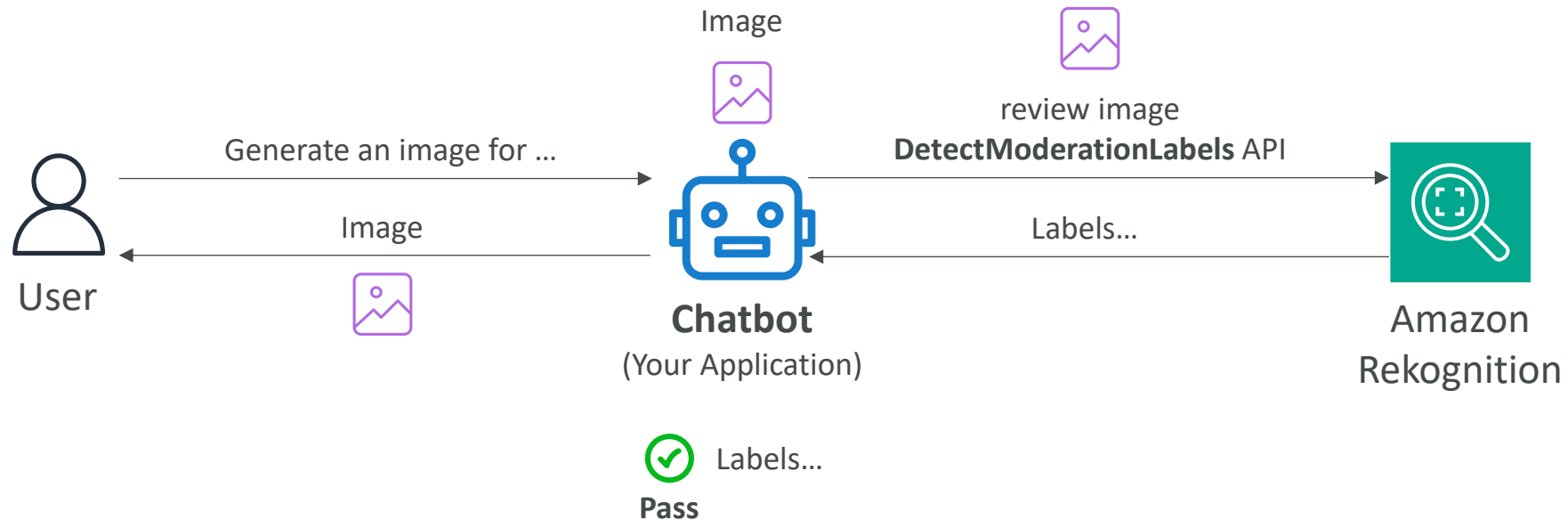


Amazon Rekognition – Content Moderation

- Automatically detect inappropriate, unwanted, or offensive content
- Example: filter out harmful images in social media, broadcast media, advertising...
- Bring down human review to 1-5% of total content volume
- Integrated with Amazon Augmented AI (Amazon A2I) for human review
- **Custom Moderation Adaptors**
 - Extends Rekognition capabilities by providing your own **labeled** set of images
 - Enhances the accuracy of Content Moderation or create a specific use case of Moderation

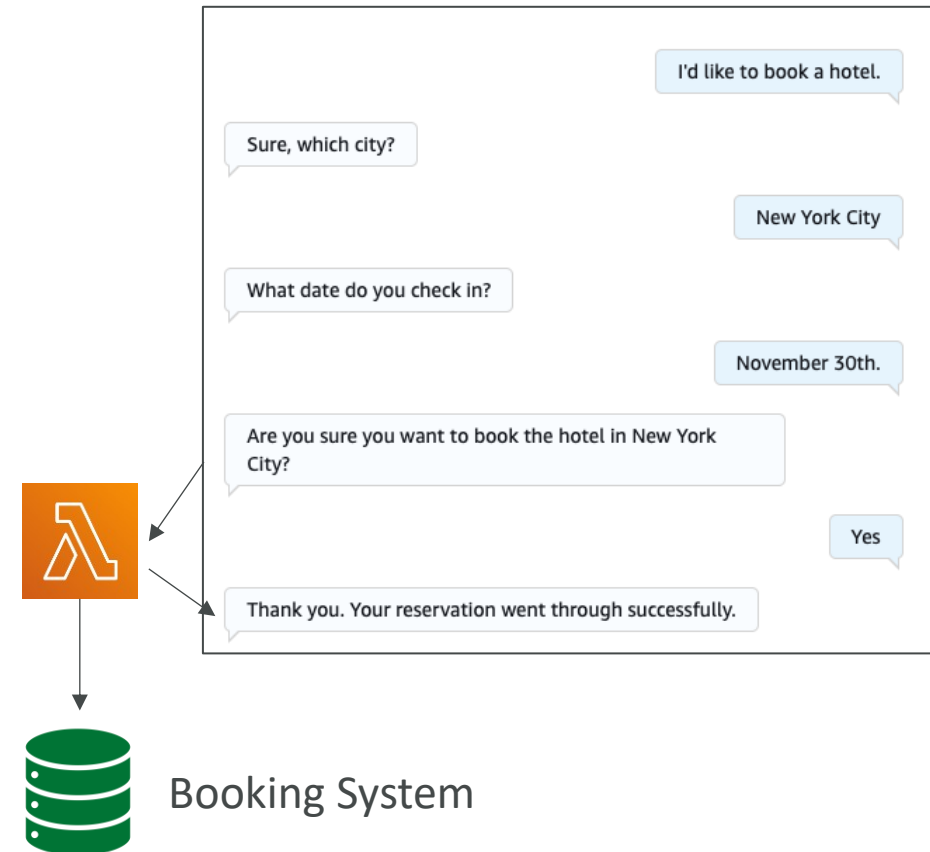


Content Moderation API – Diagram



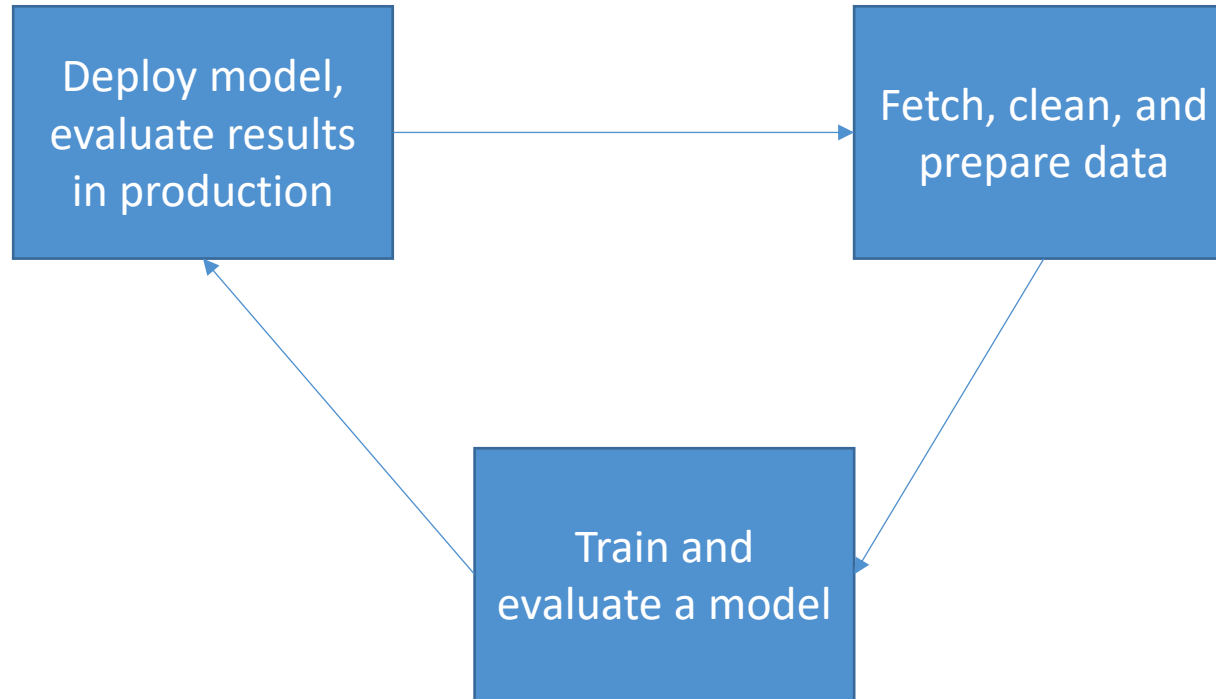
Amazon Lex

- Build chatbots quickly for your applications using voice and text
- Example: a chatbot that allows your customers to order pizzas or book a hotel
- Supports multiple languages
- Integration with AWS Lambda, Connect, Comprehend, Kendra
- The bot automatically understands the user intent to invoke the correct Lambda function to “fulfill the intent”
- The bot will ask for “Slots” (input parameters) if necessary

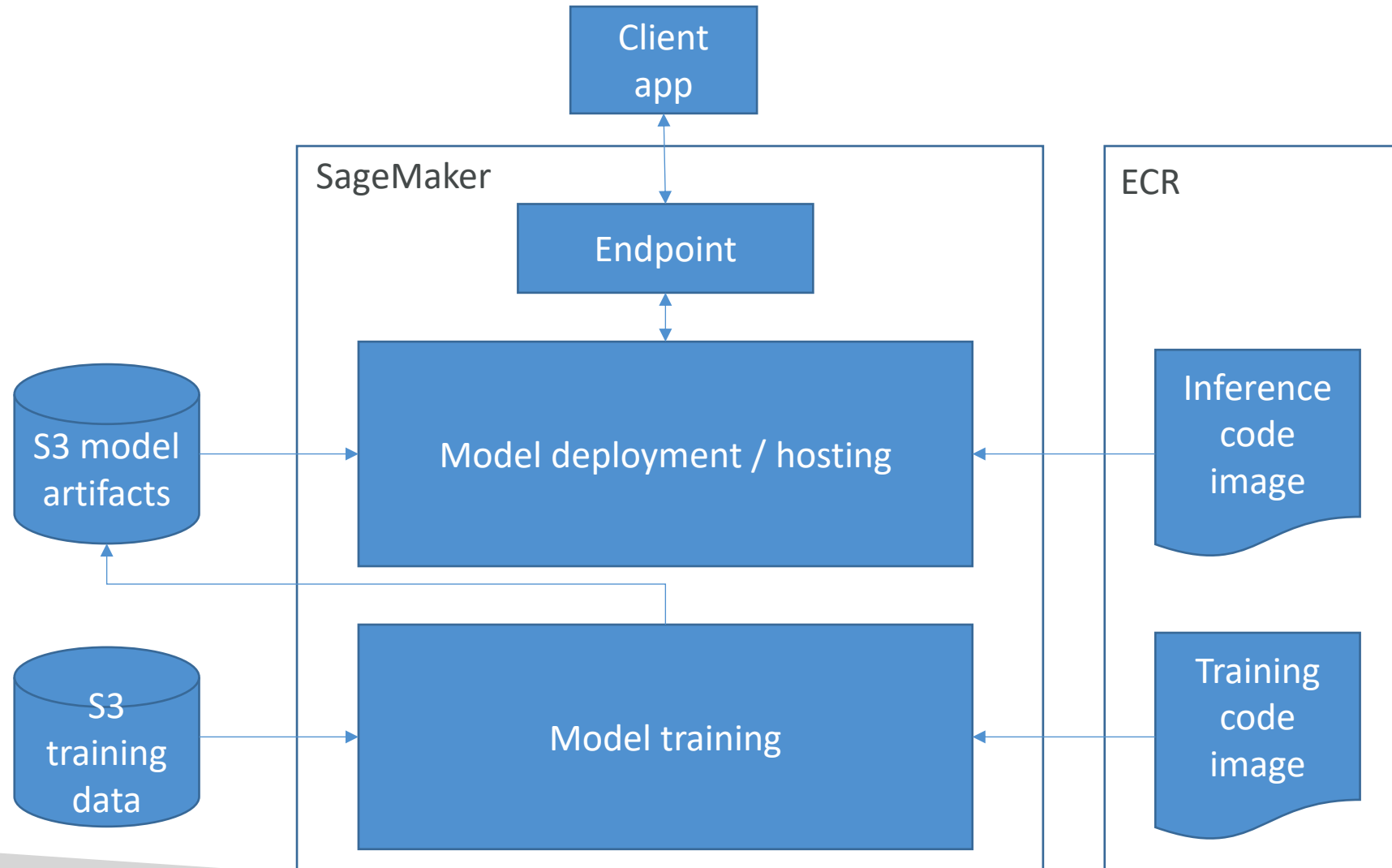


Amazon SageMaker

SageMaker is built to handle the entire machine learning workflow.

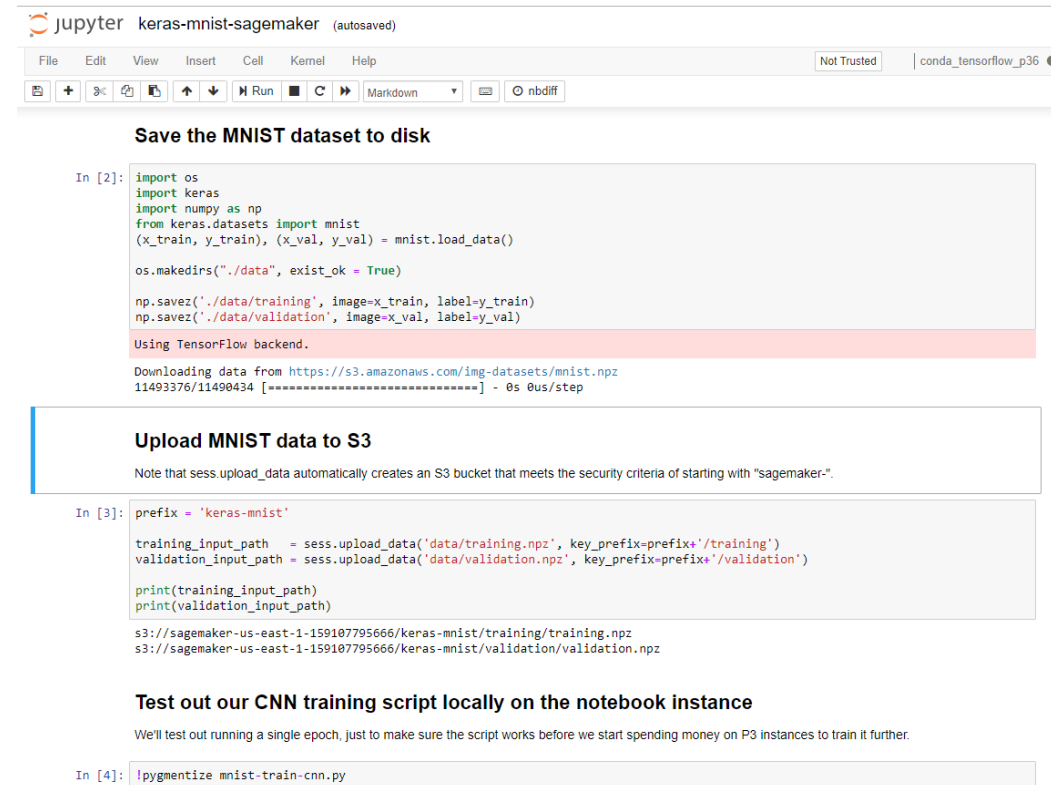


SageMaker Training & Deployment



SageMaker Notebooks can direct the process

- Notebook Instances on EC2 are spun up from the console
 - S3 data access
 - Scikit_learn, Spark, Tensorflow
 - Wide variety of built-in models
 - Ability to spin up training instances
 - Ability to deploy trained models for making predictions at scale



The screenshot shows a Jupyter Notebook interface with the title 'keras-mnist-sagemaker (autosaved)'. The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Help) and a toolbar with icons for file operations and execution. The notebook content is divided into sections:

- Save the MNIST dataset to disk**: This section contains a code cell (In [2]:) that imports necessary libraries (os, keras, numpy, mnist) and saves the MNIST dataset to local files. The code includes comments and a status bar indicating the use of the TensorFlow backend and the download progress of the dataset from S3.
- Upload MNIST data to S3**: This section contains a code cell (In [3]:) that uploads the saved training and validation data to an S3 bucket. The code uses the boto3 library to interact with S3. Below the code, the output shows the S3 paths for the uploaded files.
- Test out our CNN training script locally on the notebook instance**: This section contains a code cell (In [4]:) that runs a command to execute a Python script using Pygmentize.

So can the SageMaker console

The screenshot displays the Amazon SageMaker console interface. The top navigation bar includes the AWS logo, 'Services', 'Resource Groups', and user information (Frank Kane, N. Virginia, Support). The left sidebar shows the 'Amazon SageMaker' menu with options like Dashboard, Search, Ground Truth, Notebook, Training, and Inference. The 'Training' section is expanded, showing 'Algorithms', 'Training jobs' (highlighted), and 'Hyperparameter tuning jobs'. The main content area is titled 'Training jobs' and features a search bar, an 'Actions' dropdown, and a 'Create training job' button. Below these is a table listing training jobs with columns for Name, Creation time, Duration, and Status. All jobs shown are 'Completed'.

Name	Creation time	Duration	Status
sagemaker-tensorflow-190926-1513-010-6f1bf0d5	Sep 26, 2019 15:30 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-009-53c7332e	Sep 26, 2019 15:30 UTC	3 minutes	Completed
sagemaker-tensorflow-190926-1513-008-a53612ca	Sep 26, 2019 15:26 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-007-b06a4d5b	Sep 26, 2019 15:26 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-006-167e1e3c	Sep 26, 2019 15:22 UTC	3 minutes	Completed
sagemaker-tensorflow-190926-1513-005-fd89504c	Sep 26, 2019 15:21 UTC	5 minutes	Completed
sagemaker-tensorflow-190926-1513-004-4a31fc9c	Sep 26, 2019 15:17 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-003-187d6ee7	Sep 26, 2019 15:17 UTC	5 minutes	Completed
sagemaker-tensorflow-190926-1513-002-0fd0ad85	Sep 26, 2019 15:14 UTC	4 minutes	Completed
sagemaker-tensorflow-190926-1513-001-7f7cf798	Sep 26, 2019 15:14 UTC	3 minutes	Completed

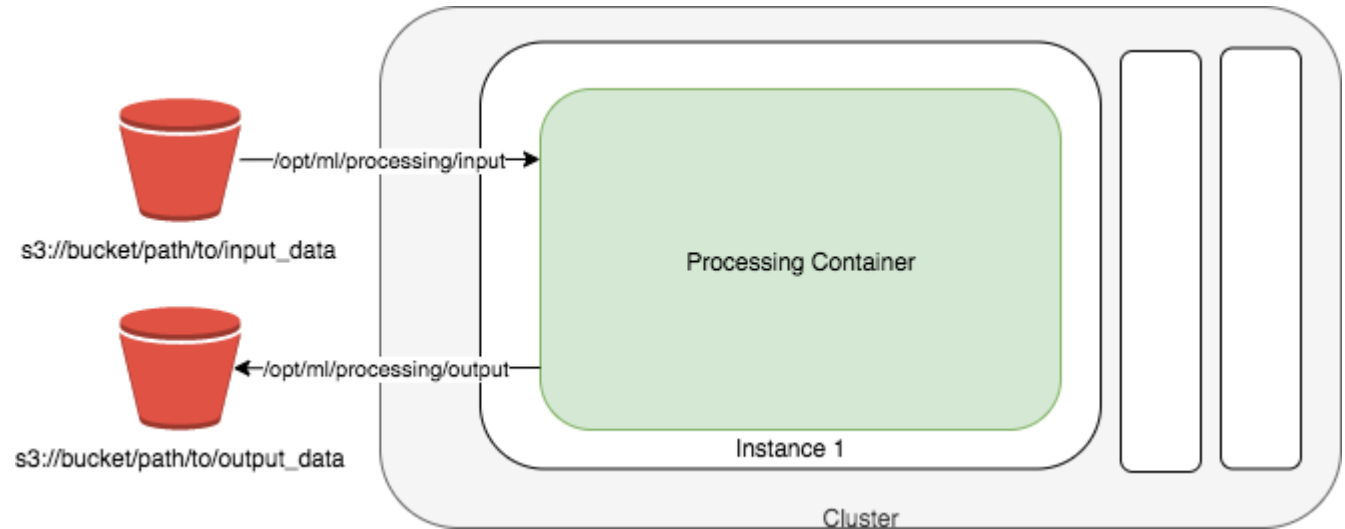
Data prep on SageMaker

- Data usually comes from S3
 - Ideal format varies with algorithm – often it is RecordIO / Protobuf
- Can also ingest from Athena, EMR, Redshift, and Amazon Keyspaces DB
- Apache Spark integrates with SageMaker
 - More on this later...
- Scikit_learn, numpy, pandas all at your disposal within a notebook



SageMaker Processing

- Processing jobs
 - Copy data from S3
 - Spin up a processing container
 - SageMaker built-in or user provided
 - Output processed data to S3



Training on SageMaker

- Create a training job
 - URL of S3 bucket with training data
 - ML compute resources
 - URL of S3 bucket for output
 - ECR path to training code
- Training options
 - Built-in training algorithms
 - Spark MLlib
 - Custom Python Tensorflow / MXNet code
 - PyTorch, Scikit-Learn, RLEstimator
 - XGBoost, Hugging Face, Chainer
 - Your own Docker image
 - Algorithm purchased from AWS marketplace



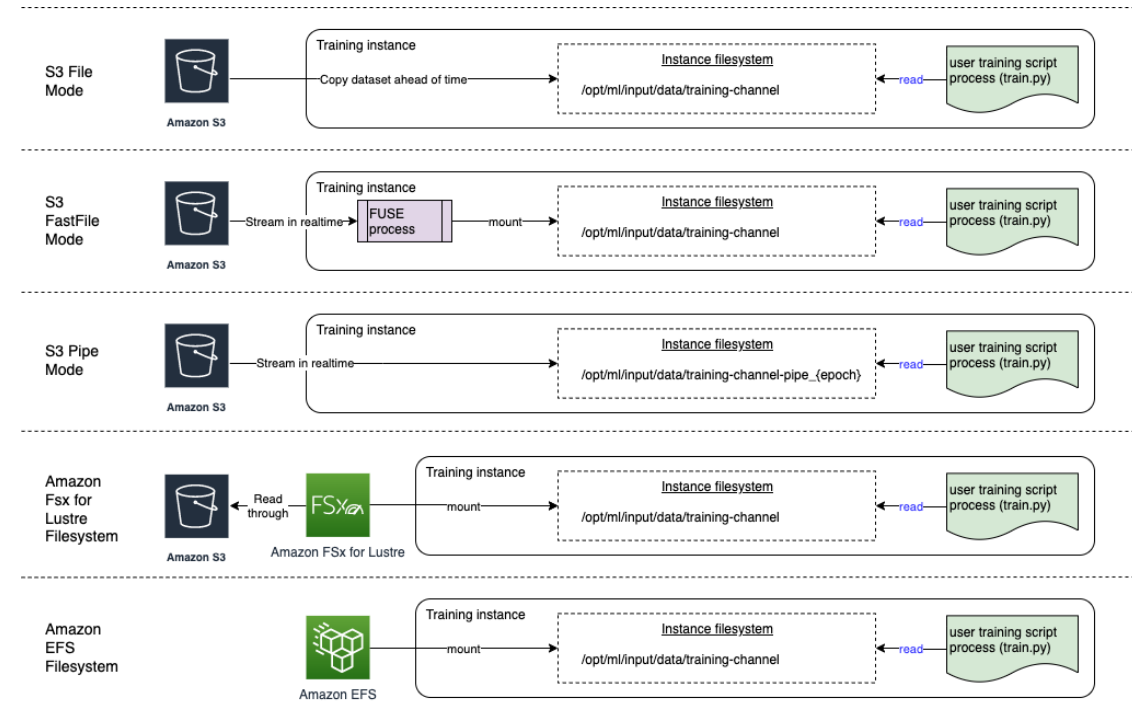
Deploying Trained Models

- Save your trained model to S3
- Can deploy two ways:
 - Persistent endpoint for making individual predictions on demand
 - SageMaker Batch Transform to get predictions for an entire dataset
- Lots of cool options
 - Inference Pipelines for more complex processing
 - SageMaker Neo for deploying to edge devices
 - Elastic Inference for accelerating deep learning models
 - Automatic scaling (increase # of endpoints as needed)
 - Shadow Testing evaluates new models against currently deployed model to catch errors



SageMaker Input Modes

- S3 File Mode
 - Default; **copies** training data from S3 to local directory in Docker container
- S3 Fast File Mode
 - Akin to “pipe mode”; training can begin without waiting to download data
 - Can do random access, but works best with sequential access
- Pipe Mode
 - Streams data directly from S3
 - Mainly replaced by Fast File
- Amazon S3 Express One Zone
 - High-performance storage class in one AZ
 - Works with file, fast file, and pipe modes
- Amazon FSx for Lustre
 - Scales to 100's of GB of throughput and millions of IOPS with low latency
 - Single AZ, requires VPC
- Amazon EFS
 - Requires data to be in EFS already, requires VPC



<https://docs.aws.amazon.com/sagemaker/latest/dg/model-access-training-data.html>